DISTRICT PANCHAYATH KASARAGOD

EQIP 2026

Educational Quality Improvement Programme for class -X

MATHEMATICS

ENGLISH MEDIUM

Standard -X

QUESTION POOL



DISTRICT INSTITUTE OF EDUCATION AND TRAINING (DIET)
KASARAGOD

EQIP 2025-26

Chief Co-ordinators

Sri. Madhusudhanan T.V.
Deputy Director of Education, Kasaragod
&
Dr. Raghurama Bhat K.
Principal, DIET Kasaragod

Resource Team

Santhoshkumar K.
HST Mathematics
GTHS Cheruvathur

Muneer M.
HST Mathematics
MPSGVHSS Bellikoth

Dileep K.V.
HST Mathematics
Durga HSS Kanhangad

Biju P.V.
HST Mathematics
GHSS Mogralputhur

Sareesh V.
HST Mathematics
GHSS Periye

Ragesh K.V.
HST Mathematics
GHSS Parappa

Academic Co-ordinator

Sri. Gireesh Babu A.
Lecturer, DIET Kasaragod



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ആശംസ

വികേന്ദ്രീകൃത ആസൂത്രണത്തിലൂടെയും നിർവ്വഹണത്തിലൂടെയും കേരളത്തിലെ ആരോഗ്യവിദ്യാഭ്യാസ മേഖലകളെ ദേശീയ തലത്തിൽ ഒന്നാമതെത്തിക്കാൻ നമുക്ക് കഴിഞ്ഞിട്ടുണ്ട്. ഈ നേട്ടങ്ങൾ കൈവരിക്കാൻ പ്രാദേശിക ഭരണകൂടങ്ങൾ സ്തുത്യർ ഹമായ പങ്കുവഹിച്ചു. ദേശീയ സംസ്ഥാനതല പഠനങ്ങൾ നമ്മുടെ കുട്ടികളുടെ പഠന നിലവാരം ഇനിയും ഉയരേണ്ടതുണ്ട് എന്ന സൂചനയാണ് നൽകുന്നത്. പത്താം ക്ലാസിലെ കുട്ടികളുടെ പഠനപ്രശ്നങ്ങൾ പരിഹരിക്കാൻ കഴിഞ്ഞ കുറെ വർഷങ്ങളായി കാസറഗോഡ് ജില്ലാ പഞ്ചായത്തിന്റെ നേതൃത്വത്തിൽ പൊതുവിദ്യാഭ്യാസ വകുപ്പും കാസറഗോഡ് ഡയറ്റും ഒത്തു ചേർന്ന് നടപ്പാക്കുന്ന **'എക്വിപ്പ്' 2025-26 (EQIP 2025-26)** പഠനപരിപോഷണ പരിപാടിക്ക് എല്ലാ പിന്തുണയും ഉറപ്പുതരുന്നു. പൊതുപരീക്ഷകളെ അഭിമുഖീകരിക്കുന്ന പത്താം ക്ലാസിലെ കുട്ടികളുടെ പഠനപ്രവർത്തനങ്ങളുടെ മികവിന്റെ അടയാളമായി മാറുകയാണ് വാർഷിക പരീക്ഷകൾ. അറിവിന്റെ തെളിമയോടെ ഓരോ വിദ്യാർത്ഥിക്കും പരീക്ഷ എഴുതാൻ കഴിയണം. വിദ്യാർത്ഥികളുടെ ജീവിതത്തിലെ ഏറ്റവും പ്രധാനപ്പെട്ട പരീക്ഷകൾക്ക് വേണ്ടി തയ്യാറാക്കിയ ചോദ്യബാങ്കിന് എല്ലാവിധ ആശംസകളും നേരുന്നു. നന്നായി പഠിക്കുക. പരീക്ഷയെ സധൈര്യം നേരിടുക. തളരാതെ മുന്നോട്ട്. വിജയം നിങ്ങളോടൊപ്പമുണ്ട്. ആശംസകൾ.

ശ്രീമതി ബേബി ബാലകൃഷ്ണൻ
ജില്ലാ പഞ്ചായത്ത് പ്രസിഡന്റ്
കാസറഗോഡ്



ആശംസ

കാസറഗോഡ് ജില്ലാ പഞ്ചായത്തിന്റെ നേതൃത്വത്തിൽ ജില്ലയിലെ പൊതുവിദ്യാഭ്യാസ മേഖലയെ ശക്തിപ്പെടുത്തുന്നതിന് നിരവധി പ്രവർത്തനങ്ങളാണ് നടന്നുവരുന്നത്. പൊതുവിദ്യാഭ്യാസ മേഖലയെ പൂർവ്വാധികം കരുത്തോടെ നാം മുന്നോട്ട് നയിക്കുകയാണ്. ഈ ഘട്ടത്തിലാണ് കാസറഗോഡ് ജില്ലാ പഞ്ചായത്തും, പൊതുവിദ്യാഭ്യാസ വകുപ്പും, വിദ്യാഭ്യാസ പരിശീലന കേന്ദ്രവും **(DIET)** പത്താം ക്ലാസ് വിദ്യാർത്ഥികളുടെ പഠന വിടവുകൾ പരിഹരിക്കുന്നതിനും ആത്മവിശ്വാസത്തോടെ പൊതുപരീക്ഷയെ നേരിടാൻ അവരെ പ്രാപ്തരാക്കുന്നതിനും വേണ്ടി പഠനപരിപോഷണ സാമഗ്രി തയ്യാറാക്കിയിട്ടുള്ളത്. ഈ പദ്ധതിയുടെ ഭാഗമായി ആത്യന്തിക വിലയിരുത്തലിനെ അഭിമുഖീകരിക്കാൻ കുട്ടികളെ സഹായിക്കുന്ന തരത്തിൽ വിവിധ വിഷയങ്ങളിൽ വിലയിരുത്തൽ സമീപനത്തിനനുസരിച്ചുള്ള ചോദ്യശേഖരമാണ് **(Question Pool)** ഈ അക്കാദമിക വർഷം തയ്യാറാക്കിയിട്ടുള്ളത്. കുട്ടികളുടെ അക്കാദമിക് മികവ് ഉറപ്പുവരുത്തിക്കൊണ്ട് മികച്ച ഗ്രേഡുകൾ നേടാൻ അവരെ സജ്ജമാക്കാൻ **എക്വിപ് 2025-26 (EQIP 2025-26)** എന്ന പേരിൽ തയ്യാറാക്കിയ ഈ പദ്ധതിക്ക് കഴിയട്ടെയെന്ന് ആശംസിക്കുന്നു.

സ്നേഹപൂർവ്വം

അഡ്വ. സരിത എസ്.എൻ.

ആരോഗ്യ-വിദ്യാഭ്യാസ സ്ഥിരം സമിതി അധ്യക്ഷ,

ജില്ലാ പഞ്ചായത്ത്, കാസറഗോഡ്



ആശംസ

ജില്ലയിലെ അക്കാദമിക പ്രവർത്തനങ്ങളെ ഏകോപിപ്പിച്ച് മുന്നോട്ട് നയിക്കുന്നതിന്റെ പ്രധാന ഉത്തരവാദിത്തം കാസറഗോഡ് ജില്ലാ വിദ്യാഭ്യാസ പരിശീലന കേന്ദ്രത്തിന്റെ (DIET) നേതൃത്വത്തിൽ കാലങ്ങളായി ചെയ്തുവരുന്നുണ്ട്. പ്രീ-പ്രൈമറി തലം മുതൽ ഹയർസെക്കണ്ടറി തലംവരെയുള്ള മേഖലകളിൽ വിവിധങ്ങളായ പദ്ധതികൾ ആസൂത്രണം ചെയ്യാനും നിർവഹിക്കാനും ഡയറ്റിന് ഇതിനകം കഴിഞ്ഞിട്ടുണ്ട്. പത്താം ക്ലാസ്സിലെ കുട്ടികളുടെ പഠനപ്രശ്നങ്ങൾ പരിഹരിക്കാൻ കഴിഞ്ഞ കുറച്ചു വർഷങ്ങളായി ഡയറ്റിന്റെ നേതൃത്വത്തിൽ വിദ്യാഭ്യാസ വകുപ്പ് നടപ്പിലാക്കുന്ന പദ്ധതിയാണ് **എക്വിപ് (Educational Quality Improvement Programme).** അതാതു വർഷത്തെ കുട്ടികളുടെ പഠനപ്രശ്നങ്ങൾ പരിഗണിച്ചുകൊണ്ടാണ് പ്രവർത്തനങ്ങൾ ആസൂത്രണം ചെയ്യുന്നത്. പത്താംതരം പാഠപുസ്തകവും വിലയിരുത്തൽ രീതിയും മാറിയ ഘട്ടത്തിൽ ആണല്ലോ നമ്മൾ. അതുകൊണ്ട് ഈ പദ്ധതിയുടെ ഭാഗമായി പത്താംതരത്തിലെ ആത്യന്തിക വിലയിരുത്തലിനെ അഭിമുഖീകരിക്കാൻ കുട്ടികളെ സഹായിക്കുന്ന തരത്തിൽ വിവിധ വിഷയങ്ങളിൽ പുതിയ വിലയിരുത്തൽ സമീപനത്തിനനുസരിച്ച് ചോദ്യശേഖരം (Question Pool) പരിചയപ്പെടുത്താനാണ് ഇപ്രാവശ്യം ഉദ്ദേശിക്കുന്നത്. ജില്ലാ പഞ്ചായത്തിന്റെ നേതൃത്വത്തിൽ ഡയറ്റിന്റെ അക്കാദമിക സഹായത്തോടെ മലയാളത്തിലും ഇംഗ്ലീഷിലും കന്നഡയിലും പത്താം തരത്തിലെ ഐ.ടി. ഒഴികെയുള്ള വിഷയങ്ങളിൽ ചോദ്യശേഖരം തയ്യാറാക്കി നൽകാനാണ് ഉദ്ദേശിക്കുന്നത്. ഈ അധ്യയനവർഷം പത്താംതരം പരീക്ഷ എഴുതുന്ന മുഴുവൻ കുട്ടികൾക്കും ഈ പദ്ധതിയുടെ പ്രയോജനം ലഭിക്കുമെന്ന് പ്രതീക്ഷിക്കുന്നു. വ്യത്യസ്ത പഠനശേഷിയും പഠനമികവുമുള്ള എല്ലാ വിഭാഗം കുട്ടികൾക്കും ഈ സാമഗ്രി പ്രയോജനപ്പെടട്ടെ എന്ന് ആശംസിക്കുന്നു. ഒരു കൂട്ടം അധ്യാപകരുടെ ആത്മാർത്ഥമായ പ്രവർത്തനമാണ് ഈ സാമഗ്രി വികസിപ്പിക്കാൻ സഹായകമായത്. അധ്യാപകരുടെ ആത്മാർത്ഥമായ പിന്തുണയും പ്രോത്സാഹനവും അനിവാര്യമായ ഈ ഉദ്യമത്തിൽ എല്ലാവരുടെയും സഹായസഹകരണങ്ങൾ തുടർന്നും പ്രതീക്ഷിച്ചുകൊണ്ട് ഏവർക്കും വിജയാശംസകൾ നേരുന്നു. ആശംസകളോടെ,

ഡോ. രഘുരാമ ഭട്ട് കെ.
പ്രിൻസിപ്പാൾ
ഡയറ്റ് കാസറഗോഡ്



ആമുഖം

കാസറഗോഡ് ജില്ലാ പഞ്ചായത്തിന്റെ നേതൃത്വത്തിൽ ജില്ലയിലെ പൊതുവിദ്യാഭ്യാസം ശക്തിപ്പെടുത്തുന്നതിന് വ്യത്യസ്തങ്ങളായ നിരവധി പ്രവർത്തനങ്ങൾ നടന്നുവരികയാണ്. അതേസമയം ദേശീയ-സംസ്ഥാന പഠനങ്ങൾ നമ്മുടെ ജില്ലയിലെ കുട്ടികളുടെ പ്രകടനം ഇനിയും മെച്ചപ്പെടേണ്ടതുണ്ട് എന്ന സൂചനയാണ് നൽകുന്നത്. ഈ പശ്ചാത്തലത്തിലാണ് ജില്ലയിലെ പൊതുവിദ്യാലയങ്ങളിൽ നിന്ന് **2025-26** അധ്യയന വർഷം എസ്.എസ്.എൽ.സി. പരീക്ഷകൾ അഭിമുഖീകരിക്കുന്ന കുട്ടികൾക്ക് പഠനപിന്തുണ നൽകുന്നതിന് ജില്ലാ പഞ്ചായത്തിന്റെയും പൊതുവിദ്യാഭ്യാസ വകുപ്പിന്റെയും സംയുക്താഭിമുഖ്യത്തിൽ വ്യത്യസ്ത വിഷയങ്ങളിൽ പഠനസാമഗ്രികൾ തയ്യാറാക്കേണ്ടതിന്റെ ആവശ്യകത ജില്ലാതല ഉന്നതാധികാര യോഗങ്ങളിൽ ചർച്ചചെയ്യപ്പെട്ടത്. ഇതിന്റെ അടിസ്ഥാനത്തിൽ ഡയറ്റ് കാസറഗോഡിന്റെ അക്കാദമിക നേതൃത്വത്തിൽ ജില്ലയിലെ മികച്ച അധ്യാപകരെ ഉൾപ്പെടുത്തിക്കൊണ്ട് പത്താംതരത്തിൽ ഐ.ടി. ഒഴിച്ചുള്ള എല്ലാ വിഷയങ്ങളിൽ വിലയിരുത്തൽ സമീപനത്തിലനുസരിച്ച് ചോദ്യശേഖരം **(Question Pool)** തയ്യാറാക്കിയിരിക്കുകയാണ്. ജില്ലയിൽ നിന്നും പൊതുപരീക്ഷയെ അഭിമുഖീകരിക്കുന്ന മുഴുവൻ എസ്.എസ്.എൽ.സി, വിദ്യാർത്ഥികൾക്കും ആത്മവിശ്വാസം വളർത്തുന്നതിനും ഉന്നത വിജയം നേടുന്നതിനും ഈ ഉദ്യമം സഹായകമാകട്ടെയെന്ന് ആത്മാർത്ഥമായി ആഗ്രഹിക്കുന്നു. ഈ പദ്ധതിയെ നെഞ്ചേറ്റിയ പ്രിയപ്പെട്ട അധ്യാപക സുഹൃത്തുക്കൾക്ക് ഈ പുസ്തകത്തെ ഫലപ്രദമായി ഉപയോഗിക്കാൻ കഴിയട്ടെ.

എല്ലാവർക്കും വിജയാശംസകൾ.

ശ്രീ. മധുസൂദനൻ ടി. വി.
ജില്ലാ വിദ്യാഭ്യാസ ഉപഡയറക്ടർ
കാസറഗോഡ്

MATHEMATICS

English Medium

Chapter 1

ARITHMETIC SEQUENCES

1. What is the comman difference of the arithmetic sequence

$2\frac{3}{4},\ 3\frac{1}{2},\ 4\frac{1}{4},\ .....$

A) $\frac{1}{2}$ B) $\frac{3}{4}$

B) $\frac{1}{4}$ C) $\frac{1}{5}$

2. The common difference of the arithmetic sequence $\sqrt{2},\ \sqrt{8},\ \sqrt{18},\ \sqrt{32},\ .....$ is,

A) $\sqrt{2},$ B) $\sqrt{6}$

C) $2\sqrt{2}$ D) $\sqrt{10}$

3. 16th term of the arithmetic sequence $5,\ 12,\ 19,\ 26,\ .....$ is

A) 100 B) 105

C) 110 D) 120

4. $2x-1,\ 2x+1,\ 5x-9$ are consecutive terms in an arithmetic sequence, what is x?

A) 1 B) 2

C) 3 D) 4

5. What is the difference between the 20th term and 15th term of the arithmetic sequence whose common difference 5?

A) 15 B) 20

C) 25 D) 30

6. If the 16th term of an arithmetic sequence is 47, what is the sum of the 31st terms?

A) 1457 B) 1456

C) 1458 D) 1459

7. If the sum of the 15th and 16th terms of an arithmetic sequence is 122, then what is the sum of the first 30 terms.

A) 1820 B) 1830

C) 1840 D) 1850

8. What is the 10th term from the last term of the arithmetic sequence 1, 5, 9, 13, ..., 201

A) 160 B) 162

C) 165 D) 168

9. The Sum of terms from 4th to 10th of an arithmetic sequence is 84,

i) What is the 7th term?

ii) What is the sum of 6th term and 8th term.

iii) What is the sum of the terms from $3r^{d}$ to 9th?

10. The 6th term of an arithmetic sequence is 25 and 10th term is 40.

i) What is the second term?

ii) What is the position of the number 55 in the sequence?

iii) What number must be added to 10th term to get 30th term?

11. The 9th term of an arithmetic sequence is 25.

i) What is the sum of the terms from the 5th to 13th term?

ii) Find the sum of first 17 term of the sequence?

12. Consider the Arithmetic sequence 4, 10, 16, ...

i) Find the common difference

ii) What is the difference of 6th term and 11th term of this sequence?

iii) Is the difference between any two terms of this sequence be 110? Why?

13. 4th term of an arithmetic sequence is 21 and 9th tem is 41.

    i) What is the common difference of this sequence?

    ii) What is the first term of this sequence?

    iii) What is the 15th term?

14. The sum of 8th term and 19th term of an arithmetic sequence is 125.

    i) Find the sum of 7th and 20th terms of this sequence.

    ii) If the 6th term is 40, find the 21st term of this sequence?

    iii) Find the sum of 26 terms of this sequence.

15. Consider the arithmetic sequence 1, 3, 5, 7, ...

    i) Find sum of first n terms of this sequence.

    ii) Find the sum of first n terms of the arithmetic sequence
    $\frac{1}{n}, \frac{3}{n}, \frac{5}{n}, \frac{7}{n}, \ldots\ldots\ldots\ldots$

    iii) Find the sum of first 2025 terms of the arithmetic sequence
    $\frac{1}{2025}, \frac{3}{2025}, \frac{5}{2025}, \ldots\ldots\ldots\ldots$

16. The sum of 4th and 6th terms of an arithmetic sequence is 120.

    i) Find the 5th term of this sequence.

    ii) Find the sum of first 9 terms of this sequence?

    iii) If the 2nd term of this sequence is 20, find the 8th term of this sequence.

Chapter 1

ARITHMETIC SEQUENCES

ANSWERS

1. B
2. A
3. C
4. D
5. C
6. A
7. B
8. C

9. i) 12
   ii) 24
   iii) 84

10. i) 10
    ii) 14th term
    iii) 75

11. i) 225
    ii) 425

12. i) 6
    ii) 30
    iii) No, 110 is not a multiple of common differences.

13. i) 4
    ii) 9
    iii) 65

14. i) 125
    ii) 85
    iii) 1625

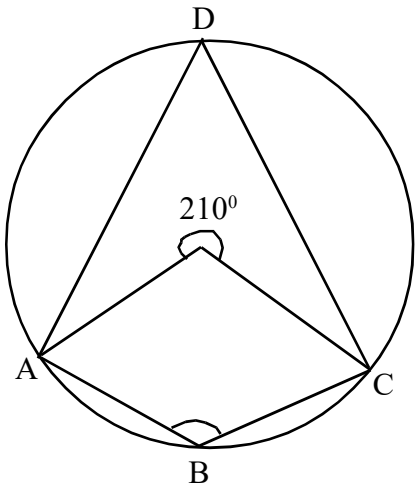
15. i) n^2
    ii) n
    iii) 2025

16. i) 60
    ii) 540
    iii) 100

Chapter 2

CIRCLES

1. In the figure central angle of an arc is 210^0. $\angle B$ and $\angle D$ are the angles on the alternate arc.



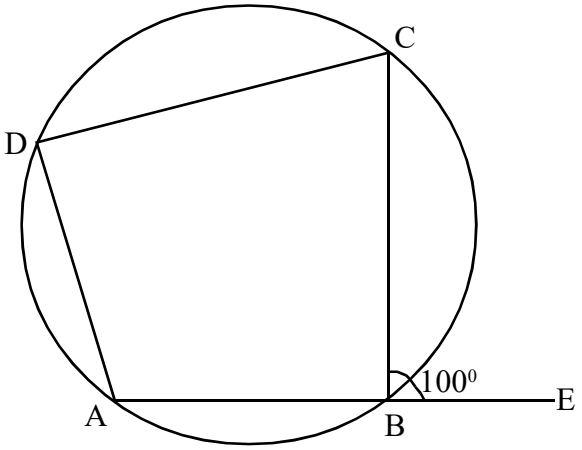
What is the measure of $\angle B$?

A) 75^0 B) 105^0

C) 210^0 D) 110^0

2. In the figure, $\angle CBE = 100^0$.

Which among the following can be $\angle A + \angle D$?



A) 200^0 B) 80^0

C) 100^0 D) 290^0

3. Which among the following is not a cyclic quadrilateral.

A) Rectangle B) Square

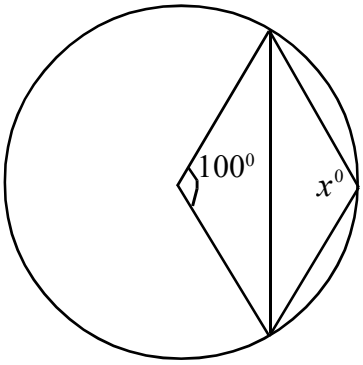
C) Isosceles Trapezium D) Parallelogram

4. In the figure O is the centre of the circle.

What is the value of x?

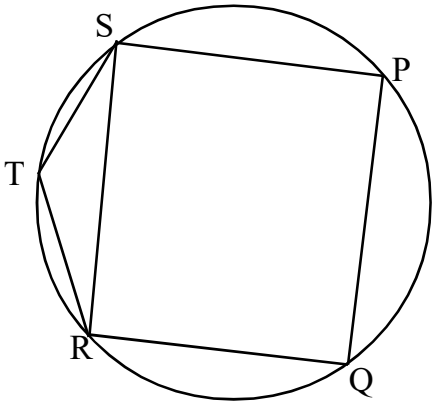
A) 130^0 B) 140^0

C) 105^0 D) 120^0



5. In the figure, PQRS is a square. What is the measure of $\angle STR$?

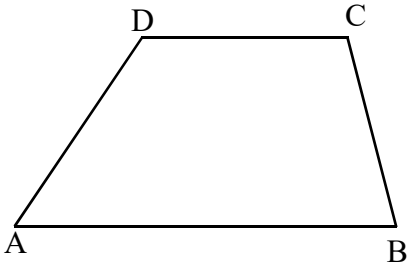
(120^0, 155^0, 135^0, 110^0)



6. Read the two statements given below.

Statement 1 : In quadrilateral ABCD the sides AB and DC are parallel and $\angle A \neq \angle B$. ABCD is a cyclic quadrilateral.

Statement 2 : Non-isosceles trapezium are not cyclic.



A) Statement 1 is true and statement 2 is false

B) Statement 1 is false and statement 2 is true

C) Both Statements are true and statement 2 is the reason of statement 1.

D) Both Statements are true and statement 2 is not the reason of statement 1

7. Read the two statements given below.

Statement 1 : Sum of outer angles of opposite vertices of a cyclic quadrilateral is 180^0.

Statement 2 : In a cyclic quadrilateral, outer angle at a vertex is equal to inner angle at the opposite vertex.

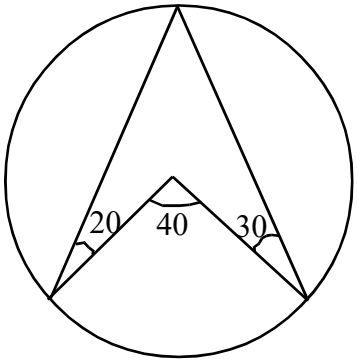
A) Statement 1 is true and statement 2 is false

B) Statement 1 is false and statement 2 is false

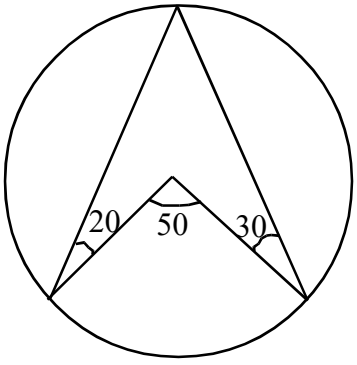
C) Both Statements are true and statement 2 is the reason of statement 1.

D) Both Statements are true and statement 2 is not the reason of statement 1

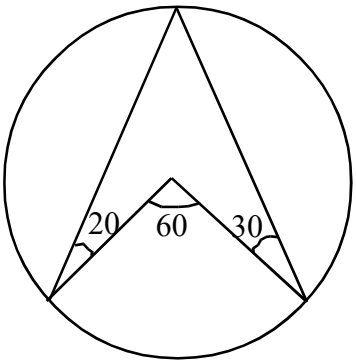
8. Which among the following figure, the angles are marked correctly?



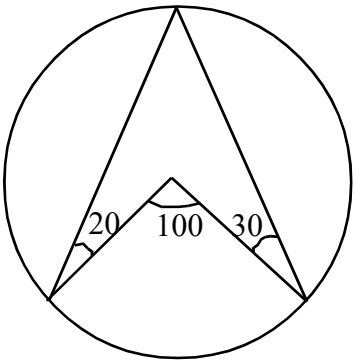
A



B



C

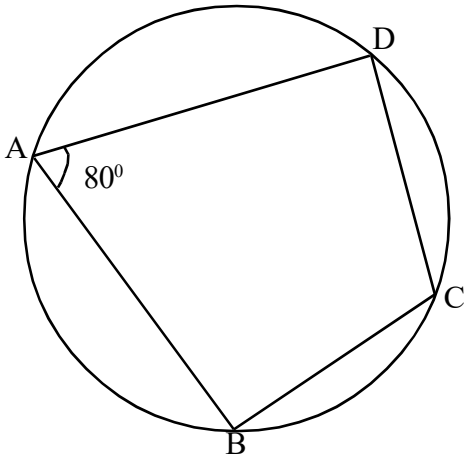


D

9. In the figure, $\angle A = 80^0$.

What is the measure of central angle of arc BAD?

(100^0, 200^0, 80^0, 160^0)



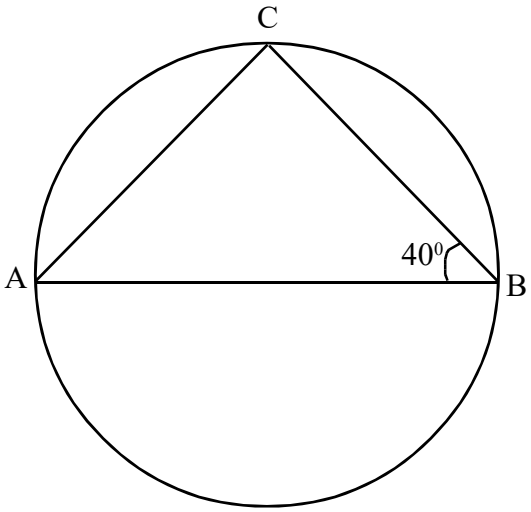
10. What is the measure of central angle of $\frac{1}{4}$ part of a circle.

(360^0, 120^0, 90^0, 60^0)

11. Read the statements below:

Statement 1 : In the figure, AB is the diameter. If $\angle B = 40^0$ then, $\angle A = 50^0$.

Statement 2 : The angle in a semicircle is a right angle.



A) Statement 1 is true and statement 2 is false

B) Statement 1 is false and statement 2 is false

C) Both Statements are true and statement 2 is the reason of statement 1.

D) Both Statements are true and statement 2 is not the reason of statement 1

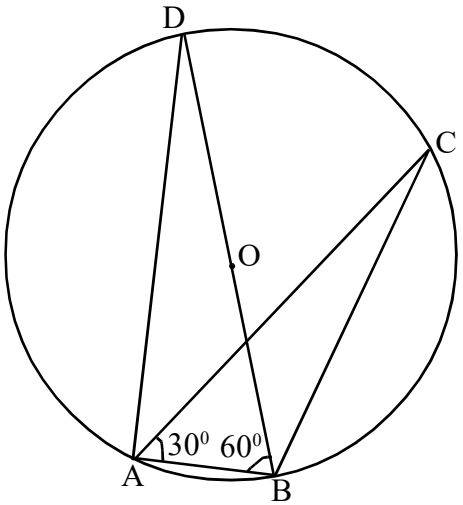
12. In the figure O is the centre of the circle.

A, B, C, D are points on the circle.

$\angle CAB = 30^0$, $\angle ABD = 60^0$

i) $\angle ADB$ = ...

ii) $\angle ABC$ = ...



13. Angles on the alternate segments of a circle are in the ratio 1:3.

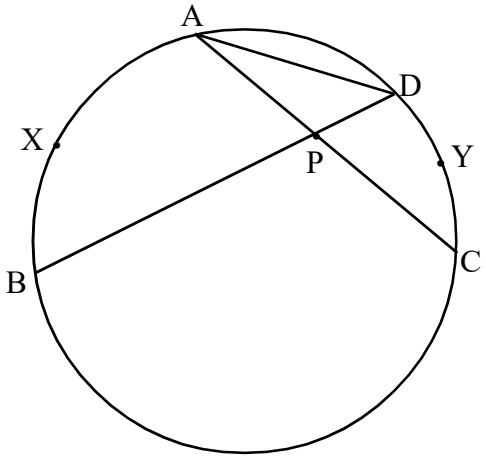
What is the measure of angles in each segment?

14. In the figure, central angle of arc AXB is 60^0 and arc CYD is 70^0.

i) Find the measure of $\angle D$?

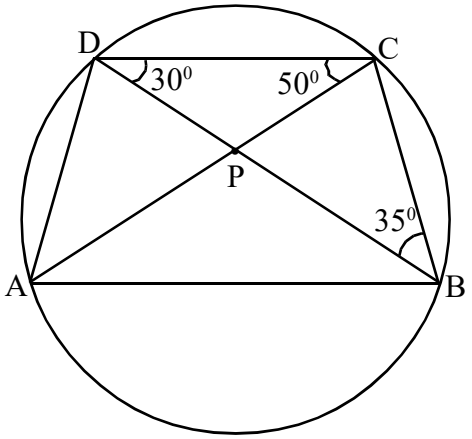
ii) Find the measure of $\angle A$?

iii) Find the measure of $\angle APB$?



15. i) Find the angles of cyclic quadrilateral ABCD in the figure.

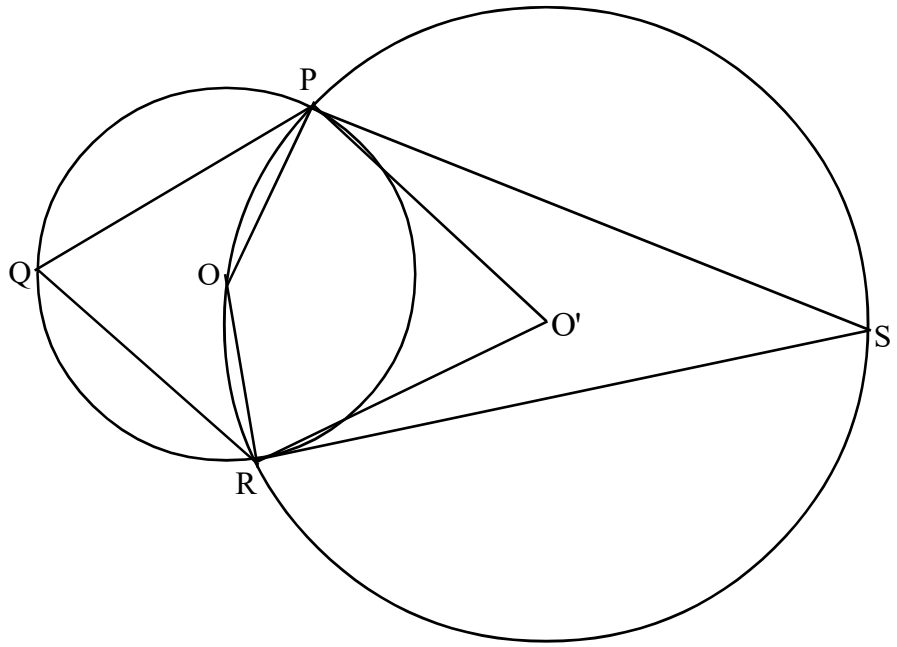
ii) Find all angles of ΔBPC ?



16. Draw a circle of any radius. Draw a chord such that angle on one segment should be three and half times the angle on the other segment.

17. Draw a triangle with circumradius 3.5 cm and two of the angles 35^0 and 72^0.

18.



In the figure, O and O' are the centre of the two circles which are intersecting at the points P and R. $\angle PQR = 80^0$.

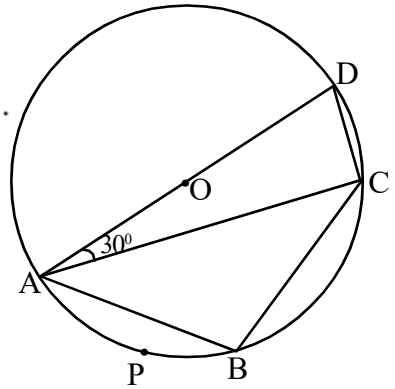
i) $\angle POR = ...$

ii) $\angle PSR = ...$

iii) $\angle PO'R = ...$

19. In the figure AD is the diameter of the circle.

The central angle of arc APB is 80^0 and $\angle DAC = 30^0$.

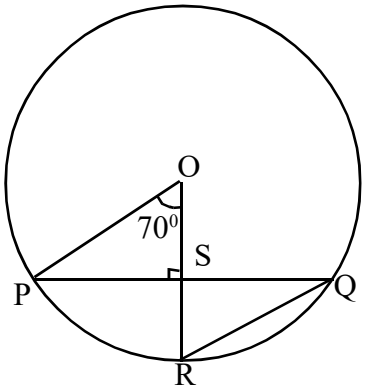


i) What is the measure of $\angle ACB$?

ii) What is the measure of $\angle ABC$?

iii) What is the measure of $\angle CAB$?

20.

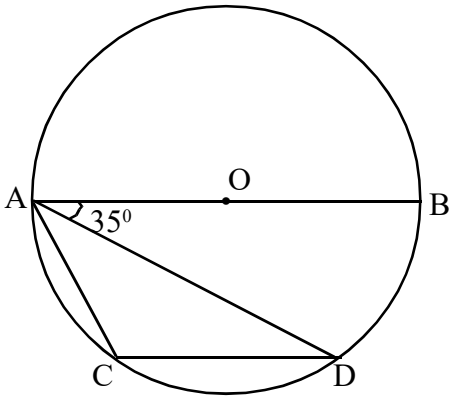


In the figure P, Q, R are the points on the circle with centre O. $\angle POR = 70^0$ and $\angle PSR = 90^0$

i) Find all the angles of ΔPOS.

ii) Find the all the angles of ΔRSQ.

21. In the figure chord CD is parallel to the diameter AB and $\angle BAD = 35^0$. Find all the angles of ΔACD.



22. In the figure A, B, C, D, E are the points on the circle with centre O and OC is parallel to DE. If $\angle OAB = 50^0$, $\angle OCB = 60^0$ and $\angle EDC = 35^0$.

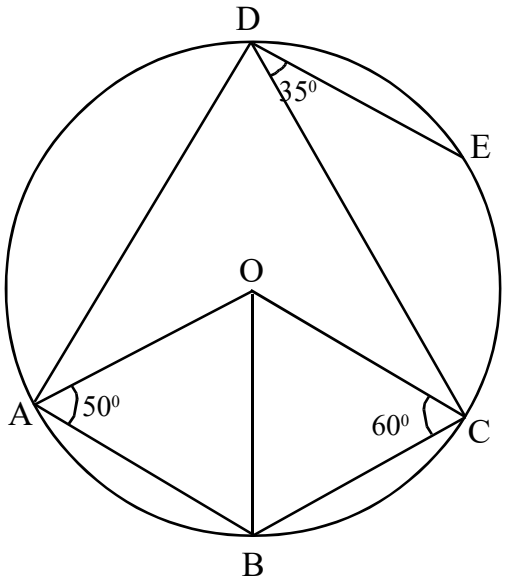
Find the measure of,

i) $\angle ABO$ = ...

ii) $\angle ADC$ = ...

iii) $\angle DAB$ = ...

iv) $\angle DAO$ = ...



23. ABCD is a cyclic quadrilateral.

$\angle A + \angle D = 210^0$ and $\angle D + \angle C = 250^0$.

i) Find $\angle A + \angle C$?

ii) Find the measure of $\angle A$ *and* $\angle C$?

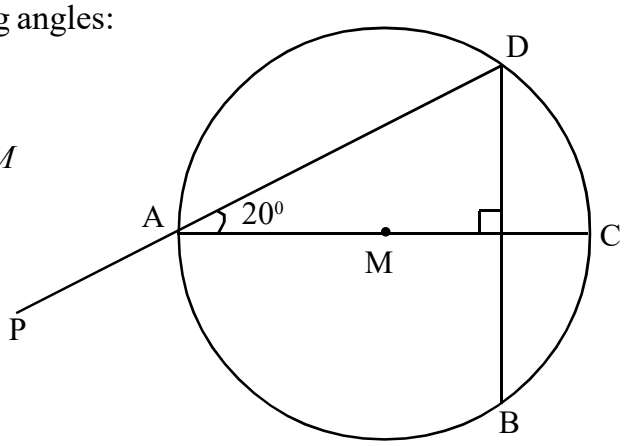
24. In the figure, Chord BD is perpendicular to diameter AC.

Find the measure of following angles:

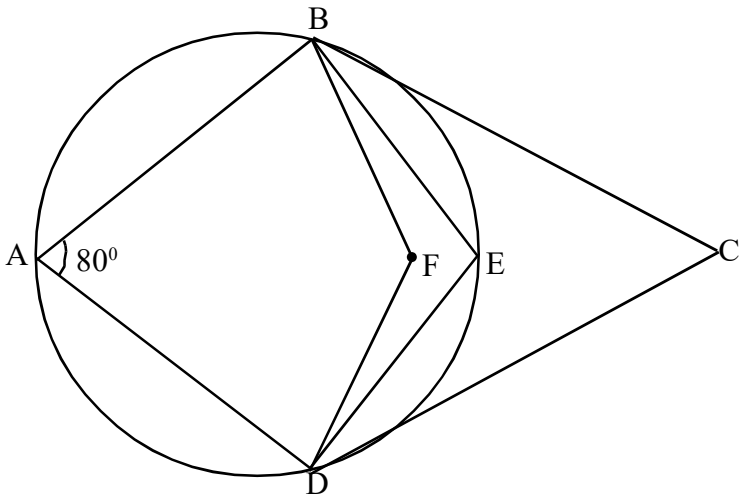
i) $\angle BAC$ ii) $\angle BCD$

iii) $\angle ADC$ iii) $\angle CDM$

v) $\angle BAP$



25. In the figure A, B, E, D are the points on the circle and $\angle A = 80^0$.



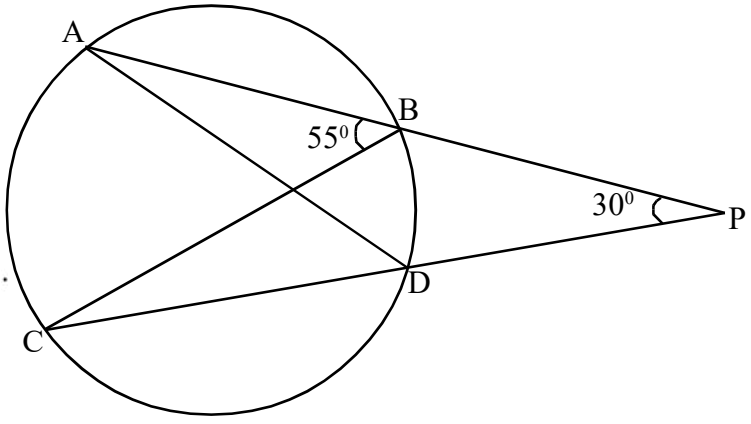
i) $\angle BED$ = ...

ii) $\angle BED + \angle BFD + \angle BCD = 270^0$

and the measure of smaller angle among these is 50^0. What is the measure of $\angle BFD$ *and* $\angle BCD$?

26. In the figure chords AB and CD extended to meet at the point P.

$\angle ABC = 55^0$ and $\angle APC = 30^0$.



i) $\angle BCD$ = ...

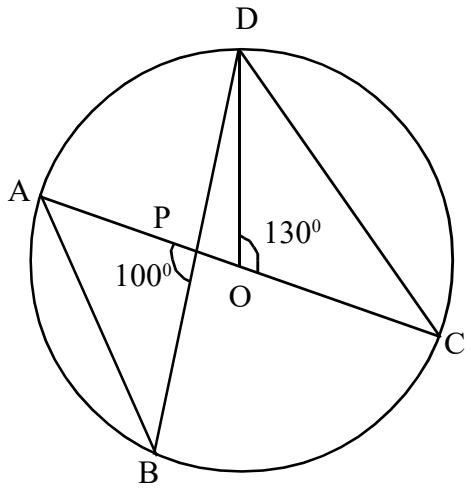
ii) $\angle BAD$ = ...

27. In the figure A, B, C, D are the points on the circle with centre O. Find the measure of following angles.

i) $\angle DBC$

ii) $\angle ACB$

iii) $\angle ABD$

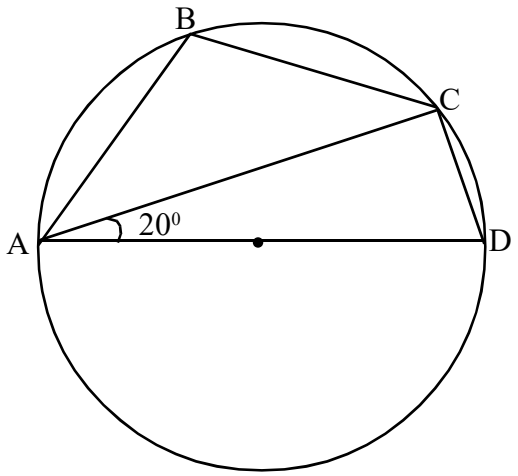


28. In the figure AD is the diameter of the circle and length of the chords AB and BC are equal.

$\angle CAD = 20^0$.

i) Find the measure of $\angle ADC$?

ii) Find the measure of $\angle BAD$?

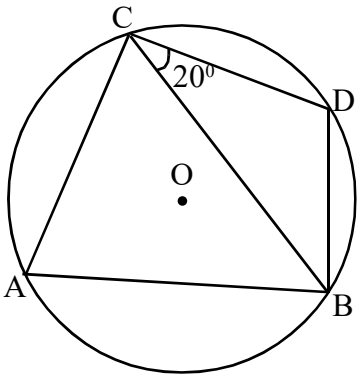


29. In the figure, ABCD is a cyclic quadrilateral and length of chord CD is equal to the radius of the circle. $\angle BCD = 20^0$

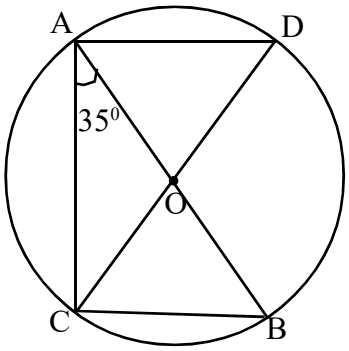
i) What is the measure of central angle of chord CD?

ii) What is the measure of $\angle CBD$?

iii) What is the measure of $\angle CAB$?



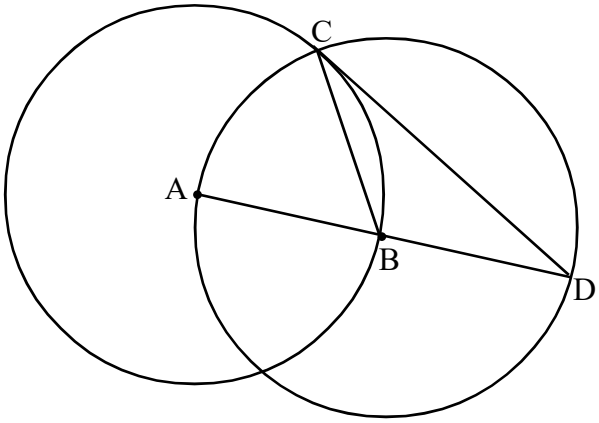
30. In the figure A, B, C, D are the points on the circle with centre O and $\angle BAC = 35^0$.



i) $\angle ACB$ = ...

ii) $\angle ADC$ = ...

31.



In the figure, A and B are the centre of two circles of same radius.

i) What is the measure of $\angle ABC$?

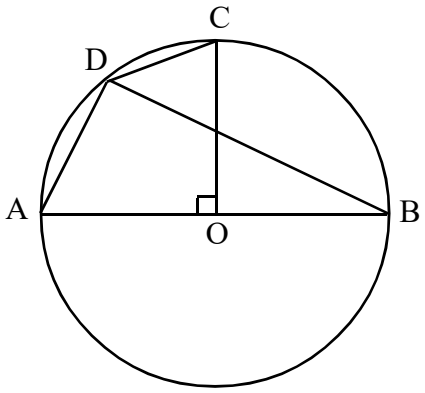
ii) What is the measure of $\angle BDC$?

32. In the figure, radius OC is perpendicular to the diameter AB.

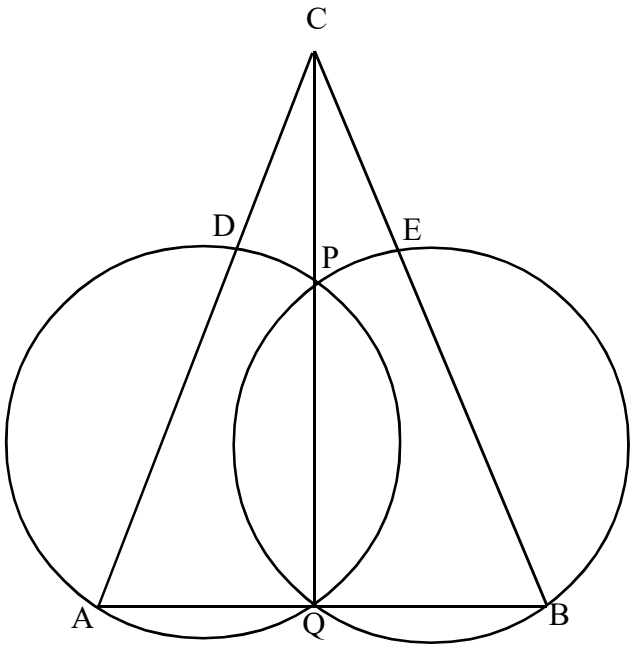
D is a point on the circle.

i) Find $\angle BDC$?

ii) Find $\angle ADC$?



33.



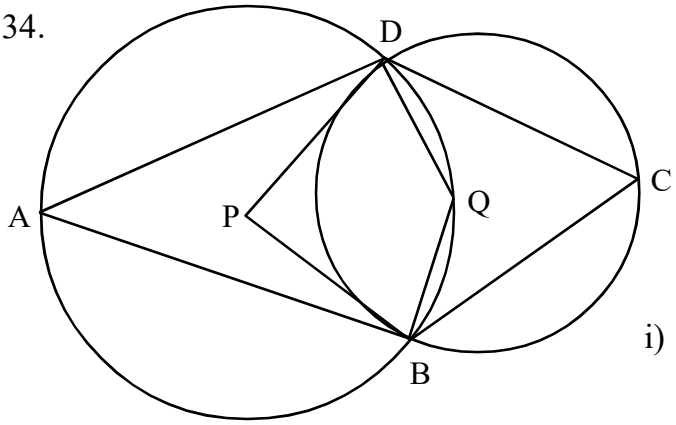
In the figure, two circles intersect at the P and Q. The chords AD and BE are extended to meet at the point C.

i) If $\angle AQP = x^0$, what is $\angle ADP$?

ii) Prove that $\angle ADP + \angle BEP = 180^0$.

iii) Prove that CDPE is a cyclic quadrilateral.

34.

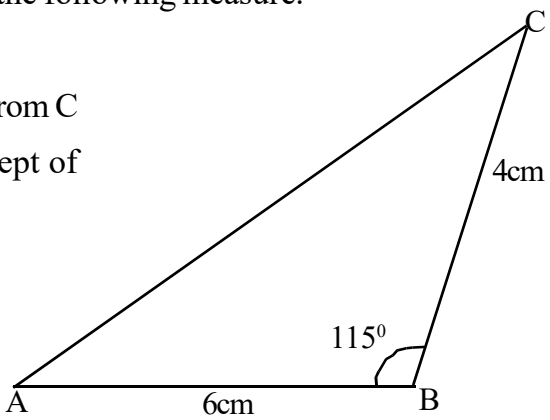


In the figure, P and Q are the centre of the two circles and $\angle BAD = 30^0$. Find the measure of the following angles.

i) $\angle BPD$ ii) $\angle BQD$ iii) $\angle BCD$

35. Draw triangle ABC with the following measure.

Draw the perpendicular from C to AB by using the concept of angle on a semi circle.

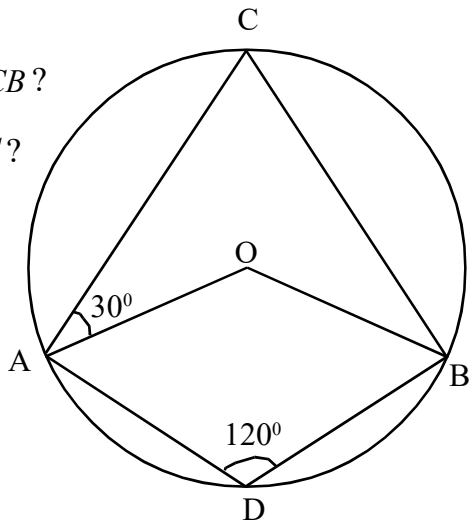


36. In the figure, O is the centre of the circle.

$\angle ADB = 120^0$, $\angle OAC = 30^0$

i) What is the measure of $\angle ACB$?

ii) Find all the angles of ΔABC ?

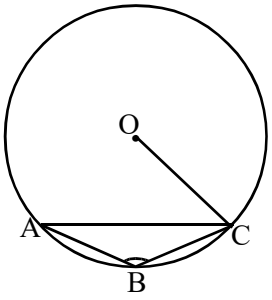


37. A, B, C are the points on a circle with centre O.

If $\angle OCA = x^0$,

i) What is $\angle OAC$?

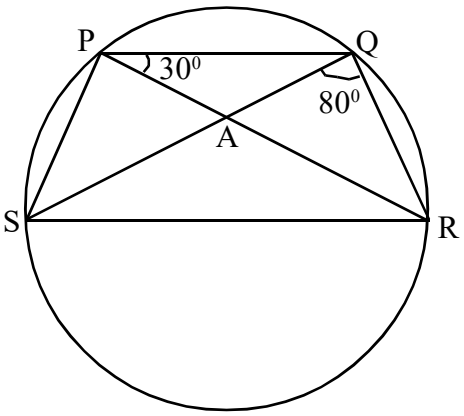
ii) Prove that $\angle ABC - \angle OCA = 90^0$?



38. Draw a triangle with circum radius 3cm and two of the angles $27½^0$ and $32½^0$?

39. In the figure, PQRS is a cyclic quadrilateral in which diagonals PR and QS intersects at A. $\angle SQR = 80^0$ and $\angle QPR = 30^0$

Find the measure of $\angle SRQ$?



Chapter 2

CIRCLES

ANSWERS

1. B
2. A
3. D
4. A
5. 135^0
6. B
7. C
8. D
9. 200^0
10. 90^0
11. C
12. i) $\angle ADB = 30^0$

    ii) $\angle ABC = 120^0$
13. Sum of angles in alternative segment is 180^0

    Angle in a segment $= \frac{1}{4} \times 180^0 = 45^0$

    Angles in alternative angle $= \frac{3}{4} \times 180^0 = 135^0$
14. i) $\angle D = \frac{60^0}{2} = 30^0$

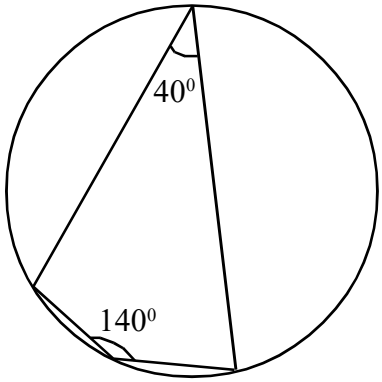
    ii) $\angle A = \frac{70^0}{2} = 35^0$

    iii) $\angle APB = \angle A + \angle D = 35^0 + 30^0 = 65^0$

15. i) $65^0,\ 85^0,\ 115^0,\ 95^0$

ii) $35^0,\ 65^0,\ 80^0$

16.



17. Construction of triangle

18. i) $\angle POR = 160^0$ ii) $\angle PSR = 20^0$ iii) $\angle PO^1R = 40^0$

19. i) $\angle ACB = 40^0$ ii) $\angle ABC = 120^0$ iii) $\angle CAB = 20^0$

20. Angles of ΔPOS are $20^0, 70^0, 90^0$

Angles of ΔRSQ are $35^0, 55^0, 90^0$

21. $\angle ADC = 35^0$, BD Joined $\angle ABD = 55^0$

$\therefore \angle ACD = 125^0$

$\angle DAC = 180^0 - (35^0 + 125^0) = 180^0 - 160^0 = 20^0$

22. i) $\angle ABO = 50^0$

ii) $\angle ADC = 70^0$

iii) $\angle DAB = 85^0$

iv) $\angle DAO = 35^0$

23. $\angle C - \angle A = 40^0$

i) $\angle C + \angle A = 180^0$

ii) $\angle C = \frac{180+40}{2} = \frac{220}{2} = 110^0$

$\angle A = \frac{180-40}{2} = \frac{140}{2} = 70^0$

24. i) $\angle BAC = 20^0$

ii) $\angle BCD = 140^0$

iii) $\angle ADC = 90^0$

iv) $\angle CDM = 70^0$

v) $\angle BAP = 140^0$

25. i) $\angle BED = 100^0$

ii) $\angle BFD = 120^0$, $\angle BCD = 50^0$

26. i) $\angle BCD = 25^0$

ii) $\angle BAD = 25^0$

27. i) $\angle DBC = \frac{1}{2}\angle DOC = \frac{130}{2} = 65^0$

ii) $\angle ACB = 100 - \angle DBC = 100^0 - 65^0 = 35^0$

iii) $\angle ABD = 25^0$

28. i) $\angle ADC = 70^0$

ii) $\angle BAD = 55^0$

29. i) $\angle COD = 60^0$

ii) $\angle CBD = \frac{1}{2}\angle COD = 30^0$

iii) $\angle CAB = 50^0$

30. i) $\angle ACB = 90^0$

ii) $\angle ADC = \angle ABC = 55^0$

31. i) $\angle ABC = 60^0$

ii) $\angle BDC = \frac{1}{2}\angle ABC = 30^0$

32. i) $\angle BDC = \frac{1}{2}\angle BOC = 45^0$

ii) $\angle ADC = \angle ADB + \angle BDC = 90^0 + 45^0 = 135^0$

33. i) $\angle ADP = 180 - x^0$

34. i) $\angle BPD = 60^0$ ii) $\angle BQD = 150^0$ iii) $\angle BCD = 75^0$

35. Construction of triangle

36. i) $\angle ACB = 60^0$

ii) 60^0 , 60^0 , 60^0

37. i) $\angle OAC = x^0$

38. Construction of triangle

39. $\angle SRQ = 70^0$

Chapter 3

ARITHMETIC SEQUENCE AND ALGEBRA

Section A

1. The sum of the first n-terms of an arithmetic sequence is $3n^2+6n$, , what is the algebraic form of the sequence?

   A) 6n+3 B) 6n-3

   C) 3n+3 D) 3n-3

2. Statement 1 : If the algebraic form of an arithmetic sequence is 6n+3, then the sum of the first 20 terms of the sequence is 1320.

   Statement 2 : If the algebraic form of the arithmetic sequence is an+b then the sum of the first 'n' terms is $\frac{an(n+1)}{2}+bn$.

   A) Statement 1 is correct, statement 2 is wrong

   B) Statement 1 is wrong, statement 2 is correct

   C) Both the statements are true, statement 2 is the correct explanation of statement 1

   D) Both the statements are true, statement 2 is not the correct explanation of statement 1

3. Statement 1 : If the algebraic form of the sum of an arithmetic sequence is $2n^2+n$, then the algebraic form of the sequence is $x_n = 4n+1$.

   Statement 2 : The algebraic form of the sum of an arithmetic sequence with $x_n = an+b$ is Pn^2+qn where $P=\frac{a}{2}$, $q=\frac{a}{2}+b$.

   A) Statement 1 is correct, statement 2 is wrong

   B) Statement 1 is wrong, statement 2 is correct

C) Both the statements are true, statement 2 is the correct explanation of statement 1

D) Both the statements are true, statement 2 is not the correct explanation of statement 1

4. Statement 1 : The sum of the first 20 terms of the arithmetic sequence 3, 6, 9, ... is 630.

   Statement 2 : The sum of first n multiples of 3 is $\frac{3n(n+1)}{2}$

A) Statement 1 is correct, statement 2 is wrong

B) Statement 1 is wrong, statement 2 is correct

C) Both the statements are true, statement 2 is the correct explanation of statement 1

D) Both the statements are true, statement 2 is not the correct explanation of statement 1

Section B

1. 63, 65, 67, ..., and 3, 10, 17, ... are two arthmetic sequences. Can any term the terms of these sequences in the same position are equal? If so write the term and position?

2. In the arithmetic sequence 10, 16, 22, ... Can the sum of any two terms of this sequence be a term of this same sequence?

3. How many three digit numbers are there, which are multiples of 7?

4. If 17th term of an arithmetic sequence is 7 more than the 10th term? Find the common difference of this sequence?

5. The sum of first 9 terms of an arithmetic sequence is 261, sum of next 6 terms is 444.

   a) Write the 5th term

b) Write the 8th term

c) Write the algebriac form of the sequence

d) Write the algebriac form of the sum of the sequence.

6. Find the following sums.

i) $1 + 2 + 3 + \ldots\ldots\ldots\ldots + 20$

ii) $4 + 8 + 12 + \ldots\ldots\ldots\ldots + 80$

iii) $7 + 11 + 15 + \ldots\ldots\ldots\ldots 83$

iv) $10 + 17 + 24 + \ldots\ldots\ldots\ldots + 143$.

7. The first term of an arithmetic sequence is 8, the common difference of this sequence is 4.

i) Write the algebriac form of the sequence.

ii) Write the algebriac form of the sum of the sequence.

8. The algebriac form of the sum of an arithmetic sequence is $2n^2 + 4n$.

i) Find the first term of this sequence

ii) Find the common difference of this sequence

iii) How many terms of this sequence starting from the first is to be added to get the sum 240?

9. Consider the arithmetic sequence 3, 8, 13, 18, ...

i) What is remainder obtained when, the terms of this sequence are divided by 5?

ii) Is 503 a term of this sequence?

iii) Write the algebraic form of this sequence?

10. i) Write the algebraic form of the arithmetic sequence 6, 10, 14, ...

ii) Write the sequence of natural numbers among the terms of the arithmetic seuqnce $\frac{6}{7}, \frac{10}{7}, 2, \ldots\ldots\ldots\ldots\ldots?$

Chapter 3

ARITHMETIC SEQUENCE AND ALGEBRA

ANSWERS

Section A

1. A
2. C) Both the statements are true, statement 2 is the correct explanation of statement 1
3. B) Statement 1 is wrong, statement 2 is correct
4. C) Both the statements are true, statement 2 is the correct explanation of statement 1

Section B

1. 63, 65, 67, ...

   $x_n = 2n+61$

   3, 10, 17, ... $x_n = 7n - 4$

   In n^{th} term of two sequences are equal,

   $2n + 61 = 7n - 4$

   $5n = 65$

   $n = \frac{65}{5}$

   is n = 13

   In both sequences 13th terms are equal

   13th term = $3+(12x7)$

   = 3 + 84 = 87

2. Algebra form

$x_n = 6n + 4$

Each term of sequence is divided by 6, remainder is 4.

n^{th} term $x_n = 6n + 4$

n^{th} term $x_m = 6m + 4$

$x_n + x_m = 6(n+m) + 6 + 2$

ie, $x_n + x_m$ is divided by 6, remainder is 2.

3. First 3 digit number which is multiple of 7 = 105

Last 3 digit number which is multiple 7 = 994

$\therefore 105 + (n-1) \times 7 = 994$

$105 + 7n - 7 = 994$

$7n = 896$

$n = \frac{896}{7} = 128$

128 three digit number which is divisible by 7.

4. **d = 1**

5. a) 5th term $= \frac{261}{9} = 29$

b) Sum of 15th terms $= 261 + 444 = 705$

8th term $= \frac{705}{15} = 47$

$3d = x_8 - x_5 = 47 - 29 = 18$

$d = \frac{18}{3} = 6$

$x_1 = x_5 - 4d$

$= 29 - 24 = 5$

c) 6n-1

d) $3n^2 + 2n$

6. i) 210

ii) 840

iii) 900

iv) 900 + 840 - 210 = 1530

7. i) 4n+4

ii) $\frac{4n(n+1)}{2} + 4n = 2n^2 + 6n$

8. i) 6

ii) 4

iii) $2n^2+4n = 240$

$n^2 +2n = 120$

$(n+1)^2 = 121$

n+1 = 11

n=10

Number of terms = 10

9. i) 3

ii) Yes

iii) 5n-2

10. i) 4n+2

ii) $x_n = \frac{4n+2}{7}$

Substituting n = 3, 10, 17, ... get the perfect number

2, 6, 10, ...

Chapter 4
MATHEMATICS OF CHANCE

1. A basket contains 18 ripen mangoes and some are not ripen. If a mango is taken from basket, without looking it the probability of getting a ripen mango is $\frac{3}{5}$. Then the total number of not ripen mangoes is?

   A) 20 B) 10

   C) 12 D) 16

2. In a leap year, probability of 53 Friday is,

   A) $\frac{3}{7}$ B) $\frac{2}{7}$

   C) $\frac{4}{7}$ D) $\frac{5}{7}$

3. Two dice each marked with numbers 1 to 6 are rolled together. What is the probability of getting both have same number?

   A) $\frac{1}{6}$ B) $\frac{2}{6}$

   C) $\frac{3}{6}$ D) $\frac{5}{6}$

4. Natural number from 1 to 40 are written on paper slips and are put in a box. One slip is taken from the box. What is the probability of getting a multiple of 5?

   A) $\frac{1}{5}$ B) $\frac{3}{5}$

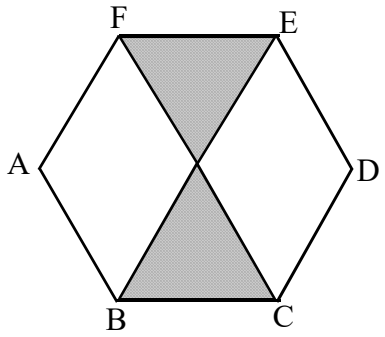
   C) $\frac{4}{5}$ D) 1

5. A box contains white balls and black balls. Total number of ball is 18. If a ball is taken from it, probability of it being black is $\frac{1}{3}$, then what is the number of white balls?

   A) 3 B) 9

   C) 6 D) 12

6.

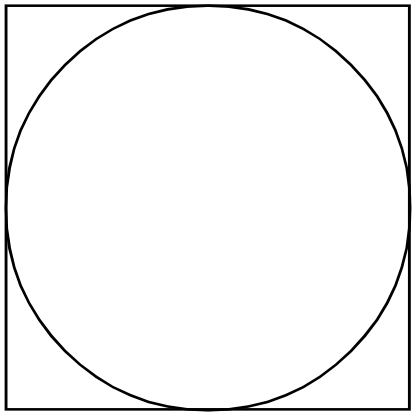


Statement 1: In the figure, ABCDEF is regular polygon. A dot is put inside the regular hexagon with out looking it. The probability that the dot to be within the shaded part is $\frac{1}{3}$.

Statement 2 : In an equilateral triangle, angles are 60^0

A) Statement 1 is true, statement 2 is false
B) Statemet 2 is true, statement 1 is false
C) Both statements are true, statement 2 is the correct reason of statement 1
D) Both statements are true, statement 2 is not the correct reason of statement 1

7.



Statement 1 : A circle is drawn inside the circle. A dot put inside the square with out looking it, the probability that the dot to be withing the circle is $\frac{\pi}{4}$.

Statement 2 : The diameter of a circle is 2cm. Its area is 4π square c.m.

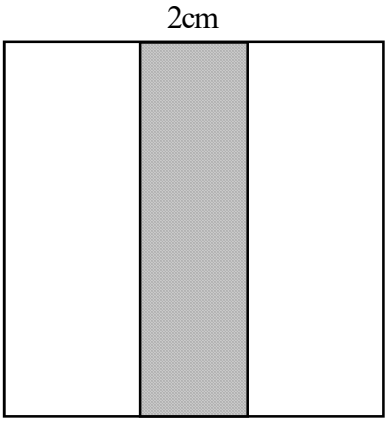
Choose the correct answer from the following.

A) Statement 1 is true, statement 2 is false
B) Statemet 2 is true, statement 1 is false

C) Both statements are true, statement 2 is the correct reason of statement 1

D) Both statements are true, statement 2 is not the correct reason of statement 1

8.



In the figure, a rectangular portion is shaded in the square of side 5cm.

Statement 1 : A dot is put inside the square without looking it, the probability that the dot lies on the shaded part and unshaded part are equal.

Statement 2 : A dot is put inside the square without looking it, the sum of the probabilities of dot being on the shaded part and unshaded part is equal to 1.

A) Statement 1 is true, statement 2 is false

B) Statemet 2 is true, statement 1 is false

C) Both statements are true, statement 2 is the correct reason of statement 1

D) Both statements are true, statement 2 is not the correct reason of statement 1

9. Statement 1 : One die with numbers 1 to 6 are rolled, then the probability of getting prime number is $\frac{1}{2}$.

Statement 2 : Prime numbers on a dice are 2, 3 and 5.

A) Statement 1 is true, statement 2 is false

B) Statemet 2 is true, statement 1 is false

C) Both statements are true, statement 2 is the correct reason of statement 1

D) Both statements are true, statement 2 is not the correct reason of statement 1

10. Statement 1 : Natural numbers from 1 to 15 are written on paper slips and are put in a box. One slip is taken from the box with out looking it, then probability of even number is $\frac{7}{15}$.

Statement 2 : Total number of even numbers between 1 to 15 is 7.

A) Statement 1 is true, statement 2 is false

B) Statemet 2 is true, statement 1 is false

C) Both statements are true, statement 2 is the correct reason of statement 1

D) Both statements are true, statement 2 is not the correct reason of statement 1

11. One asked to say a two digit number, what is the probability that the number is a perfect square?

12. Natural numbers from 1 to 30 are written on paper slips and are put in a box. One slip is taken from each box without looking.

    i) What is the probability of it being an even number?

    ii) What is the probability of it being a prime number?

13. A box contains 15 red beads and some blue beads. It we take one bead from this box without looking, then probability of getting blue bead is $\frac{4}{9}$.

    i) What is the probability of getting red bead?

    ii) Find the number of blue beads?

    iii) If we add 3 more red bead, then what is the probability of getting blue bead?

14. In a class, the total number of students is 40. One student is to be selected from the class. The probability of selecting a boy from the class is $\frac{3}{8}$.

    i) How many boys are there in the class?

    ii) How many girls are there in the class?

    iii) How many more boys should be join to the class so that the probabiity of getting a boy is $\frac{1}{2}$.

15. One is asked to say a two digit number

    i) Find the total number of two digit number.

    ii) What is the probability of both digits being the same.

    iii) What is the probability of the first digit being greater than the second?

    iv) What is the probability of the first digit being less than the second?

16. Two boxes containing natural numbers from 1 to 10. One slip is taken from each box, without looking. What is the probability that the number on the both slips being a prime number?

17. A bag contains 5 red balls and some blue balls. One ball is taken from each bag without looking. The probability of getting a blue ball is three times the probability of getting a red ball.

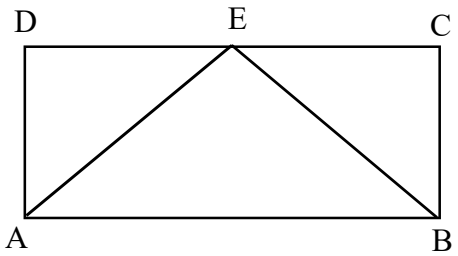
    i) If the total number of blue ball is x, then find the total number of balls.

    ii) What is the probability of getting red ball?

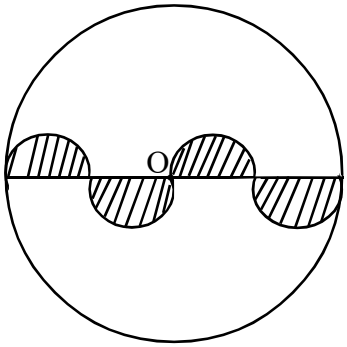
    iii) What is the probability of getting blue ball?

    iv) Find the total number of blue balls?

18. ABCD is a rectangle; E is the mid-point of CD. If we put a dot in the figure without looking it. What is the probability that it would be outside triangle AEB?



19. In the figure, all semicircles have equal areas. A dot is put inside the circle with out looking it, what is the probability that the dot lies on the shaded part



20. One is asked to say a natural numbers from 5 to 30.

i) What is the probability of it being an even number?

ii) What is the probability of it being a prime number?

21. A person is asked to say a three-digit number.

i) What is the probability that all the three digits of this number are same?

ii) What is the probability that the number is an palindrome number?

22. There are 20 boys and 25 girls in Class 10 A and 30 boys and 20 girls in Class 10B. One student is to be chosen from each class.

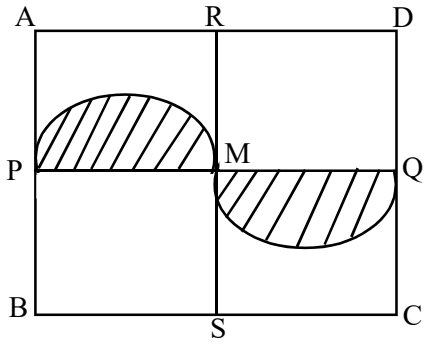
i) What is the total number of possible selections?

ii) What is the probability of both being boys?

iii) What is the probability of one being a boy and one being a girl?

iv) What is the probability of at least one being a girl?

23. In the figure, PQ, RS are the lines joining the midpoints of the sides of square ABCD. Two semicircles are drawn with PM and QM as diameters. A dot is marked inside the square with out looking it. What is the probability that the dot lies on the shaded part.



24. Two dice with colour red and black, each marked with numbers 1 to 6 are roled together.

i) If the possible numbers got from dice are written as pairs, how many pairs would be there?

ii) In how many pairs, the product of the numbers is odd?

iii) What is the probability of getting sum odd?

iv) What is the probability of getting an odd number on the red die and even number on the black die?

•

Chapter 4

MATHEMATICS OF CHANCE

ANSWERS

1. C) 12
2. B) $\frac{2}{7}$
3. A) $\frac{1}{6}$
4. A) $\frac{1}{5}$
5. D) 12
6. C) Both statements are true, statement 2 is the correct reason of statement 1
7. A) Statement 1 is true, statement 2 is false
8. B) Statemet 2 is true, statement 1 is false
9. C) Both statements are true, statement 2 is the correct reason of statement 1
10. C) Both statements are true, statement 2 is the correct reason of statement 1
11. Perfect squares

    1, 4, 9, ... 81, 100

    Two digit perfect sqaures

    16, 25, 36, 49, 64, 81

    Probability = $\frac{6}{90}$

12. i) $\frac{15}{30}$

    ii) Prime numbers

    2, 3, 5, 11, 13, 17, 19, 23, 29

    Chance = $\frac{10}{30}$

13. blue beads $= \frac{4}{9}$

i) red beads $= \frac{5}{9}$

ii) $\frac{5}{9}x = 15$

$x = \frac{15x9}{5} = 27$

blue beads = 27-15 = 12

iii) Chance $= \frac{12}{30}$

14. i) $40x\frac{3}{8} = 15$

ii) $40 - 15 = 25$

iii) 10

15. i) 90

ii) $\frac{9}{90}$

iii) $\frac{45}{90}$

iv) $\frac{36}{90}$

16. $\frac{16}{100}$

17. i) $5 + x$

ii) $\frac{5}{5+x}$

iii) $\frac{x}{5+x}$

iv) $\frac{3x5}{5+x} = \frac{x}{x+5}$

$x = 15$

No.of blue balls = 15

18. $\frac{1}{2}$

19. Consider radius of by circle 2r,

Radius of small circle is = $\frac{r}{2}$

Area of 4 semi circles = $\frac{4x\pi r^2}{8} = \frac{\pi r^2}{2}$

Area of big circles = $4\pi r^2$

Probability = $\frac{\pi r^2}{2} \div 4\pi r^2$

$= \frac{1}{8}$

20. Even numbers = 6, 8, ... 30

i) $\frac{13}{26}$

ii) Prime numbers 5, 7, 11, 13, 17, 19, 23, 29

$\frac{8}{26}$

21. i) $\frac{9}{900}$

ii)

101	201	909
111	212	919
121	222	.....
131	232	.....
.....	.....	.....
.....	.....	.....
191	292	999

} 10

$$10 \text{ x } 9 = \frac{90}{900}$$

22.

10A	10B
20B	30B
25G	20G
45	50

i) $45 \text{x} 50 = 2250$

ii) $P(BB) = \frac{20x30}{2250} = \frac{600}{2250}$

iii) $P(BG \text{ or } GB) = \frac{20x20 + 25x30}{2250} = \frac{1150}{2250}$

iv) $1 - \frac{600}{2250} = \frac{1650}{2250}$

23. Sides of squares = 4r

Area = $(4r)^2 = 16r^2$

Area of shaded region = πr^2

Probability = $\frac{\pi r^2}{16r^2} = \frac{\pi}{16}$

24. i) 36

ii) $P(OO) = \frac{3x3}{36} = \frac{9}{36}$

iii) $P(OE \text{ or } EO) = \frac{3x3 + 3x3}{36} = \frac{18}{36}$

iv) $P(OE) = \frac{3x3}{36} = \frac{9}{36}$

Chapter 5
SECOND DEGREE EQUATIONS

1. Which of the following are the solutions of the second degree equation $x^2 - 5x + 6 = 0$.

   A) 2, 3 B) -2, -3

   D) 1, 6 D) 5, -6

2. If one solution of second degree equation $x^2 - 7x + k = 0$ is 3, then what is the value of k?

   A) 10 B) 12

   C) 18 D) 21

3. $x^2 + 4x - 5 = 0$ is a second degree equation, then which of the following is a solution?

   A) 6 B) 5

   C) 1 D) -1

4. Sum of two numbers is 4 and their product is 2, then the numbers are .....

   A) $1+\sqrt{2},\ 1-\sqrt{2}$ B) $\sqrt{2}+1,\ \sqrt{2}-1$

   C) $2+\sqrt{2},\ 2-\sqrt{2}$ D) $\sqrt{2}+2,\ \sqrt{2}-2$

5. Read the following statements given below.

   Statement 1 : The sum of first 10 terms of the arithmatic sequence 99, 97, 95, is 900.

   Statement 2 : The sum of first 90 terms of the arithmatic sequence 99, 97, 95, is 900.

   Which of the following is true?

   A) Statement 1 is true, statement 2 is false

   B) Statement 1 is false, statement 2 is true

   C) Both statements are true

   D) Both statements are false

6. If x is a natural number then,
   i) Which number is to be added to $x^2 + 8x$ to get a perfect square?
   ii) If $x^2 + px + 36$ is a perfect square, then find the value of p?
   iii) If $x^2 + px + q$ is a perfect square, then prove that $p^2 = 4q$.

7. The product of 2 consecutive mutiples of 6 is 352.
   i) By taking one multiple of 6 as x, then write the given statement as an algebraic expression.
   ii) Find the numbers

8. The sum of consecutive natural numbers starting from 1 is 465. How many natural numbers were added?

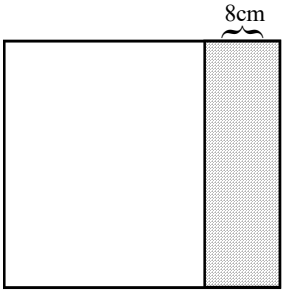
9. The sum of squares of 2 consecutive natural number is 365. Find the numbers.

10. i) If x is an integer, the value of the polynomial $x^2 + 5x$ is 14. By taking x as the same number, then find the value of the polynomial $x^2 + 5x - 14$?
    ii) Find the solutions of the equation $x^2 + 5x - 14 = 0$.

11. The sum of square of a natural number and its six times is 160.
    i) By taking the natural as x , write a second degree equation?
    ii) Find the number.

12. How many terms of the arithmatic sequence 4, 6, 8, ... must be added to get 460?

13. From a square, a rectangle of width 8 cm is cut out. The area of the remaining rectangle is 84 cm^2.
    i) By taking the length of side of the square as x, write the length of the side of rectangle.
    ii) Find the length of sides of the square.



14. How many terms of the arithmatic sequence 59,57,55,..... must be added to get the sum 500?

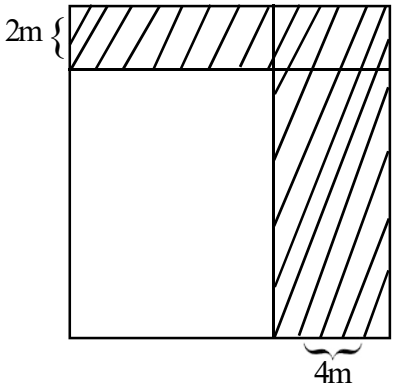
15. A wire of length 80 cm is bent to make a rectangle. The area of the rectangle is 384 cm^2. Calculate the length and breadth of the rectangle?

16. Keerthi is 6 years older than Anu. After 10 years the product of ages of Keerthi and Anu is 247.
    i) By taking the present age of Anu as 'x' years, write the present age of Keerthi.
    ii) Calculate the present ages of Keerthi and Anu?

17. From a rectangle of sides 25 cm & 15 cm a small square is cut out. The area of the remaining part is 326 cm^2. Calculate the length of the side of square which is cut out from the rectangle.

18. The difference of two numbers is 12 and their product is 448. Find the numbers.

19. In a square ground, two roads of width 2m and 4m are constructed as shown below. The area of the remaining part is 575m^2. Calculate the length and breadth of the ground after constructing the roads.



20. The perimeter and area of a rectangle are 80cm and 204 cm^2 respectively. Calculate the length and breadth of the rectangle?

Chapter 5

SECOND DEGREE EQUATIONS

ANSWERS

1. A

2. B

3. C

4. C

5. C

6. i) 16

   ii) 12

   iii) $\left(\frac{P}{2}\right)^2 = q$

   $P^2 = 4q$

7. i) $x\ (x+6) = 352$

   $x^2 + 6x = 352$

   ii) $x^2 + 6x + 9 = 352 + 9 = 361$

   $(x+3)^2 = 19^2$

   $x = 19 - 3 = 16$

   Numbers 16, 22

8. $1+2+3+.....+n = 465$

   $\frac{n(n+1)}{2} = 465$

   $n^2 + n = 930$

   $n^2 + n + \frac{1}{4} = 930 + \frac{1}{4}$

   $\left(n + \frac{1}{2}\right)^2 = \frac{3721}{4} = \left(\frac{61}{2}\right)^2$

$$n+\frac{1}{2}=\frac{61}{2}$$

n = 30

9. $n^2+(n+1)^2=365$

$$2n^2+2n+1=365$$

$$2n^2+2n=364$$

$$n^2+n=182$$

$$n^2+n+\frac{1}{4}=182+\frac{1}{4}$$

$$\left(n+\frac{1}{2}\right)^2=\frac{729}{4}=\left(\frac{27}{2}\right)^2$$

$$n+\frac{1}{2}=\frac{27}{2}$$

Numbers 13, 14

10. i) 0

ii) $x^2+5x-14=0$

$$(x-2)\times(x+7)=0$$

$$x=2,\ -7$$

11. i) $x^2+6x=160$

ii) $x^2+6x+9=160+9=169$

$$(x+3)^2=13^2$$

$$x=10$$

12. $x_n=2n+2$

$$S_n=\frac{n}{2}(4+2n+2)$$

$= \frac{n}{2}(2n+6)$

$= n^2 + 3n$

$n^2 + 3n = 460$

$n^2 + 3n + \frac{9}{4} = 460 + \frac{9}{4}$

$\left(n + \frac{3}{2}\right)^2 = \frac{1849}{4} = \left(\frac{43}{2}\right)^2$

$n + \frac{3}{2} = \frac{43}{2}$

$n = 20$

13. i) $x,\ x-8$

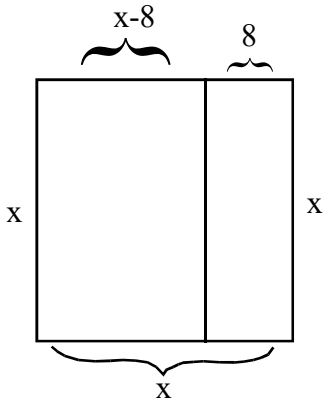
ii) $x(x-8) = 84$

$x^2 - 8x = 84$

$x^2 - 8x + 16 = 100$

$(x-4)^2 = 10^2$

$x = 14$



14. $x_n = -2n + 61$

$S_n = \frac{n}{2}\ [59 + -2n + 61]$

$= \frac{n}{2}\ [-2n + 120]$

$= -n^2 + 60\ n$

$-n^2 + 60n = 500$

$n^2 - 60n = -500$

$n^2 - 60n + 900 = -500 + 900 = 400$

$(n-30)^2 = (\pm 20)^2$

n = 20 +30 or n=30-20

= 50 = 10

15. Length = x, Breadth = 40-x

$x(40-x)=384$

$40x-x^2=384$

$x^2-40x=-384$

$x^2-40x+400=400-384=16$

$(x-20)^2=4^2$

$x=24$

Length = 24 cm Breadth = 16 cm

Or

Length = 20+x Breadth = 20-x

$(20+x)\ (20-x)=384$

$400-x^2=384$

$x^2=400-384=16$

$x=4$

Length = 24 cm, Breadth = 16 cm

16. i) $x+6$

ii) $(x+10)\ (x+16)=247$

$x^2+26x+160=247$

$x^2+26x+169=256$

$(x+13)^2=16^2$

$x=16-13=3$

Age of Anu = 3

Age of Keerthi = 9

17. Let 'x' be the side of the square

$x^2=25x15-326=49$

$x=7$

18. Numbers $x,\ x+12$

$(x+12)x = 448$

$x^2 + 12x = 448$

$x^2 + 12x + 36 = 484$

$(x+6)^2 = (\pm 22)^2$

$x = 22 - 6 = 16$ *or* $x = -22 - 6 = -28$

Numbers **16, 28**

& -28, -16

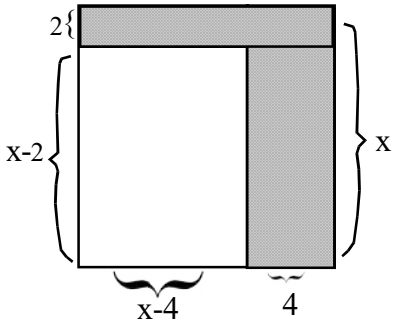
19. $(x-2)\ (x-4) = 575$

$x^2 - 6x + 8 = 575$

$x^2 - 6x + 9 = 576$

$(x-3)^2 = 24^2$

$x = 27$



Length = 25cm, Breadth = 23cm

20. 2 (Length + Breadth) = 80, Length + Breadth = 40

Length = x, Breadth = 40- x

$x\ (40-x) = 204$

$40x - x^2 = 204$

$x^2 - 40x = -204$

$x^2 - 40x + 400 = -204 + 400 = 196$

$(x-20)^2 = 14^2$

$x = 20 + 14 = 34$

Length = 34cm, Breadth = 6cm

Or

Length = 20+x, Breadth = 20-x **x = 14**

(20+x) (20-x) = 204 Length = 20 + 14 = 34 cm, Breadth = 6 cm

$400 - x^2 = 204$

$x^2 = 400 - 204 = 196$

Chapter 6
TRIGNOMETRY

1. In triangle ABC, $\angle B = 90^0$, $\text{Sin } A = \frac{3}{5}$. What is the value of CosA?

   A) $\frac{4}{5}$ B) $\frac{5}{4}$

   C) $\frac{3}{4}$ D) $\frac{2}{5}$

2. What is the circumradius of an equilateral triangle with sides 4cm?

   A) $8\sqrt{3}$ B) $\frac{8}{\sqrt{3}}$

   C) $4\sqrt{3}$ D) $\frac{4}{\sqrt{3}}$

3. A rhombus has a side of 12cm and an angle of 120^0. Which of the following is the distance between the opposite sides?

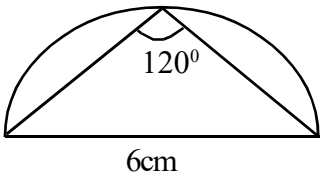
   A) 6 B) $6\sqrt{2}$

   C) $6\sqrt{3}$ D) $9\sqrt{3}$

4. I f SinA = Cos B then, A+B = ...?

   A) 30^0 B) 45^0

   C) 60^0 D) 90^0

5. 

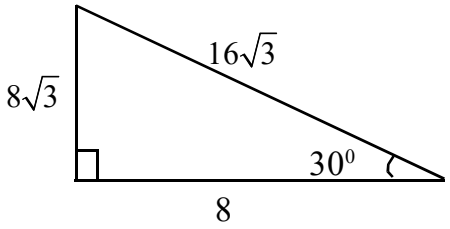
   A segment of a circle is shown in the figure. What is the radius of the circle?

   A) 3 B) $3\sqrt{3}$

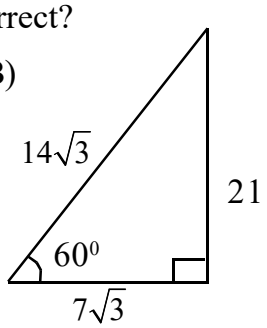
   C) $2\sqrt{3}$ D) $\frac{3}{\sqrt{3}}$

6. In the following figures which one is correct?

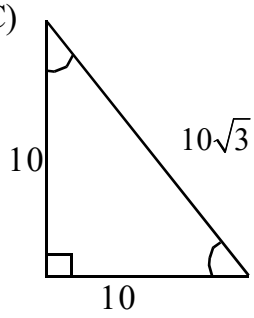
A)



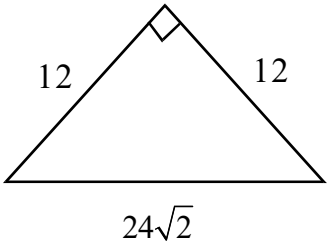
B)



C)



D)



7. Statement 1 : The circumradius of an equilateral triangle with side 3cm is $\sqrt{3}$.

   Statement 2 : The length of any chord of a circle is twice the product of the radius and the sine of half the central angle.

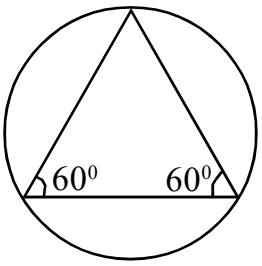
   A) Statement 1 is true and statement 2 is false
   B) Statement 1 is false and statement 2 is true
   C) Both the statements are true, statement 2 is the correct reason of statement 1
   D) Both the statements are true, statement 2 is not the correct reason of statement 1

8. Statement 1 : $(\text{Sin}60^0 \times \text{Cos}\ 30^0) + (\text{Sin}30^0 \times \text{Cos}\ 60^0) = 1$

   Statement 2 : $\text{Sin}\,45^0 = \frac{1}{\sqrt{2}},\ \text{Cos}\,45^0 = \frac{1}{\sqrt{2}}$

   A) Statement 1 is true and statement 2 is false
   B) Statement 1 is false and statement 2 is true
   C) Both the statements are true, statement 2 is the correct reasonof statement 1
   D) Both the statements are true, statement 2 is not the correct reason of statement 1

9. Statement 1 :

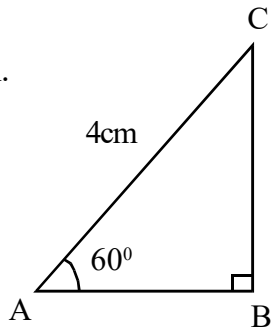


If the radius of the circle in the figure is $\sqrt{3}$ cm, then the sides of the triangle will be 3cm.

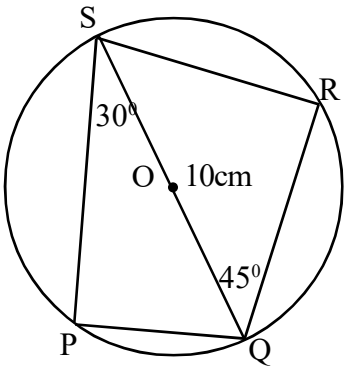
Statement 2 : The length of any chord of a circle is twice the product of the radius and the sine of half the central angle.

A) Statement 1 is true and statement 2 is false
B) Statement 1 is false and statement 2 is true
C) Both the statements are true, statement 2 is the correct reasonof statement 1
D) Both the statements are true, statement 2 is not the correct reason of statement 1

10. In the figure $\angle B = 90^0$, $\angle A = 60^0$ and AC = 4cm.
i) Find the value of $\angle C$.
ii) Find the length of BC?
iii) Calculate the area of ΔABC.

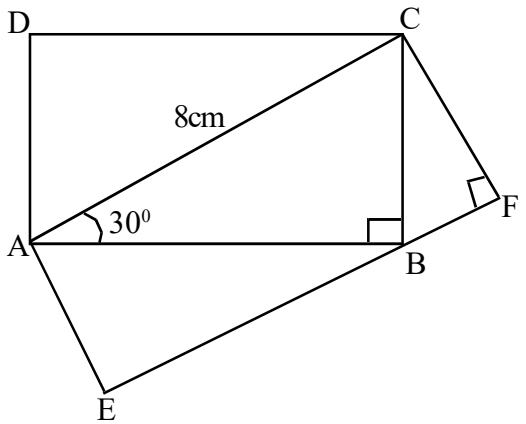


11.



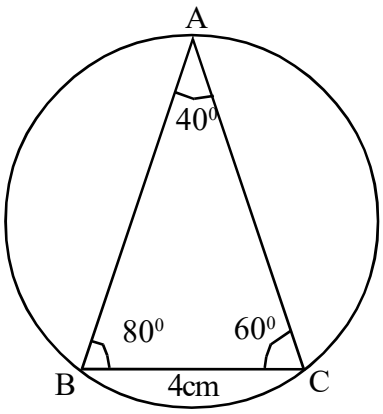
In the figure, diameter of the circle,
SQ = 10cm, $\angle PSQ = 30^0$ *and* $\angle SQR = 45^0$
i) Find the value of $\angle SPQ$.
ii) Calculate the perimeter of the quadrilateral PQRS.

12. In the figure ABCD, AEFC are rectangles.
AC = 8cm and $\angle CAB = 30^0$.
i) Find the length of BC
ii) Calculate the area of the rectangle ABCD



iii) Calculate the sum of area of the rectangles ABCD and AEFC.

13. In the figure $\angle A = 40^0$, $\angle B = 80^0$, $\angle C = 60^0$, $BC = 4cm$.

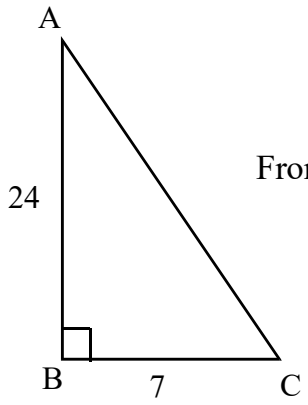


i) Find the diameter of the circle?

ii) Find out the length of other two sides of the triangle ABC

(Sin40⁰ = 0.6428, Sin 80⁰ = 0.9848)

14.



From the given triangle, find the following values.

i) SinA, CosA

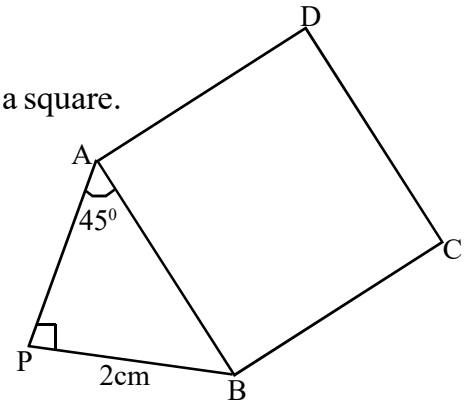
ii) SinC, CosC

15. In triangle ABC, $\angle C = 90^0$, $\angle A = 30^0$ *and* $AB = 40cm$.

Find the other two values of ΔABC.

16. In the figure, APB is a right triangle and ABCD is a square.

Also PB=2cm, $\angle PAB = 45^0$.



i) What is the length of a side of the square?

ii) Calculate the area of the square ABCD?

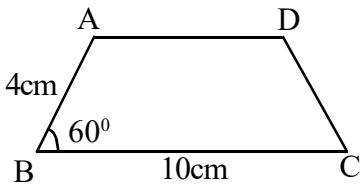
17. In ΔABC, AB = 7cm, BC = 12cm, $\angle B = 40^0$.

and Sin 40⁰ = 0.6428.

i) Find the perpendicular distance between A and BC.

ii) Calculate the area of ΔABC.

18. Calculate the area of the given isosceles trapezium ABCD.

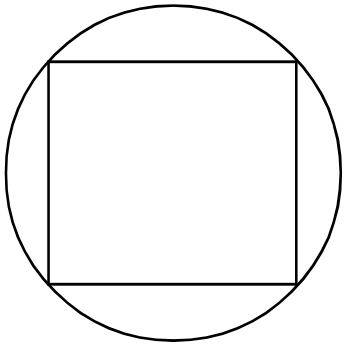


19. The perimeter of a square whose vertices all lie on a circle is 40cm. Then,

    i) What is the length of a side of the square?

    ii) What is the diameter of the circle?

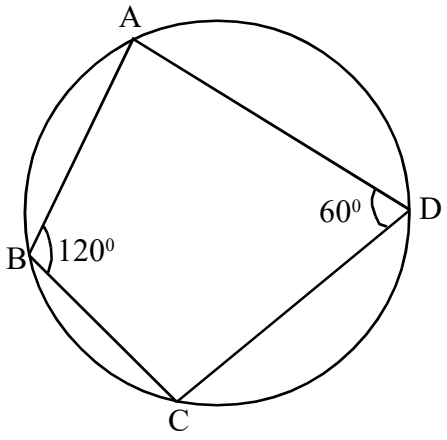
    iii) Find the perimeter of the circle.



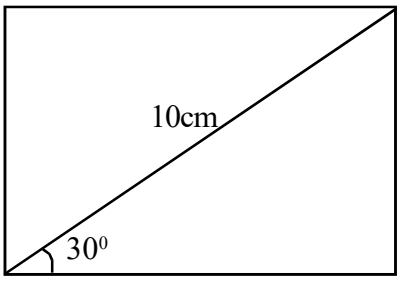
20. In Cyclic quadrilateral ABCD
    AB = BC, AD = DC.
    radius of the circle is 5cm.

    i) What is the value of $\angle ACB$?

    ii) Find the area of the quadrilateral ABCD.



21. In the figure, the length of a diagonal of the rectangle is 10cm and angle made by the longest side and diagonal is 30^0, Find the area of the rectangle.



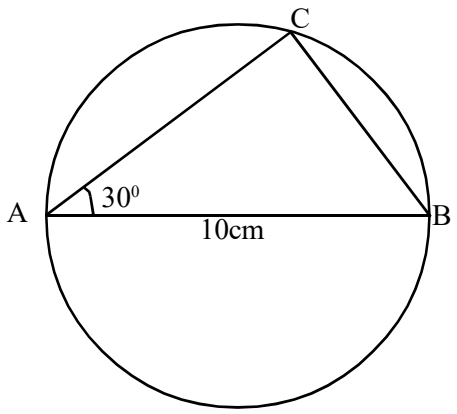
22. In the figure the length of the diameter

AB = 10cm and $\angle A = 30^0$.

i) Find the value of $\angle ACB$?

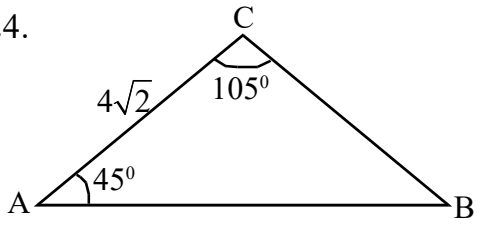
ii) What part of the diameter are the perpendicular sides of ΔABC?

iii Find the area of the ΔABC.



23. A chord is drawn for a circle of radius 4cm and the angle of a larger segment is 40^0. Find the length of the chord? (Sin 40^0 = 0.64).

24.



In ΔABC, $\angle C = 105^0$,

$\angle A = 45^0$, $AC = 4\sqrt{2}$ *cm*

i) What is the perpendicular distance from C to AB?

ii) Calculate the perimeter of ΔABC.

25. In the figure, $\angle C = 65^0$, $AB = 9cm$.

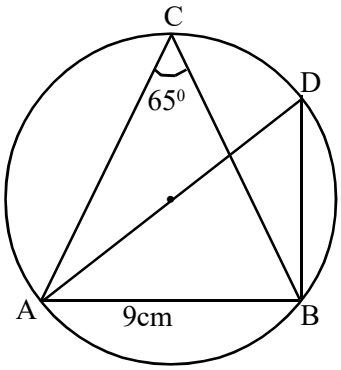
and AD is the diameter of the circle.

i) Find the value of $\angle ADB$?

ii) Calculate the circumradius of ΔABC.

iii) Find the length of the sides of the ΔABD.

(Sin65^0 = 0.91, Sin 25 = 0.42, Cos 65 = 0.42)



26. From the top of a 20m tall tower a person sees the top of the building at an angle of elevation 50^0 and its foot at an angle of depression 30^0.

i) Draw a rough figure?

ii) Find the distance between tower and buiding?

iii) Find the distance from the top of the tower to foot of the building.

iv) Find the height of the building?

(Sin 50 = 0.76)

27. A person standing on the ground sees the top of a tower at an angle of elevation of 60^0 and he move 40m towards the tower, he sees it at an elevation of 75^0.

i) Draw a rough figure.

ii) Find the height of the tower.

(Sin 75^0 = 0.9659, tan 75 = 3.7321)

28. From the top of a building a person sees to cars parked at the same side with an angle of elevation 30^0, 45^0 and distance between the cars is 50m.

i) Draw a rough figure

ii) Find the height of the building.

Chapter 6
TRIGNOMETRY
ANSWERS

1. A) $4/5$

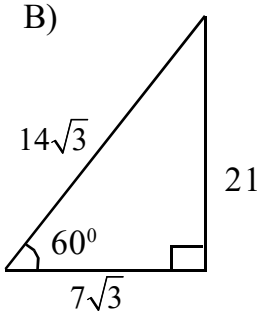
2. D) $\frac{4}{\sqrt{3}}$

3. C) $6\sqrt{3}$

4. D) 90^0

5. C) $2\sqrt{3}$

6. B)



7. C) Both the statements are true, statement 2 is the correct reason of statement 1

8. D) Both the statements are true, statement 2 is not the correct reason of statement 1

9. C) Both the statements are true, statement 2 is the correct reason of statement 1

10. i) $\angle C = 30^0$ ii) $BC = 2\sqrt{3}$ iii) $\frac{1}{2} x 2 x 2\sqrt{3} = \frac{4\sqrt{3}}{2} = 2\sqrt{3}\ cm^2$

11. i) $\angle SPQ = 90^0$

ii) $\angle PQS = 60^0,\ PQ = 5cm,\ \ PS = 5\sqrt{3}$

$SR = QR = \frac{10}{\sqrt{2}}$ $\therefore$ perimeter $= 5 + 5\sqrt{3} + 2x\frac{10}{\sqrt{2}}$

$= 5 + 5\sqrt{3} + 10\sqrt{2}$ cm

12. i) BC = 4 cm

ii) $AB = 4\sqrt{3}$, Area of Prectangle ABCD = $16\sqrt{3}\ cm^2$

iii) EF = 8cm, $CF = 2\sqrt{3}$, Area of AEFC = $16\sqrt{3}$

Sume of areas = $32\sqrt{3}$

13. i) Diameter = $\frac{4}{\text{Sin } 40} = \frac{4}{0.6428} = 6.228 \approx 6.22$

ii) $AC = 6.22 \ x \ \text{Sin } 80 = 6.13$

$AB = 6.22x\sqrt{3}/2 = 3.11x1.732 = 5.39$

14. i) $\text{Sin } A = 7/25 \quad (AC = 25)$

$\text{Cos } A = 24/25$

ii) $\text{Sin } C = 24/25$

$\text{Cos } C = 7/25$

15. $\Delta ABC \ angles \ are \ 30^0, \ 60^0, \ 90^0$

$\therefore BC = 20cm \quad AC = 20\sqrt{3}\ cm$

16. i) $2\sqrt{2}\ cm$

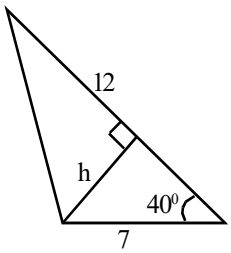
ii) $8cm^2$

17. i) $\text{Sin } 40^0 = \frac{h}{7}$

$\therefore h = 7x \text{Sin } 40$

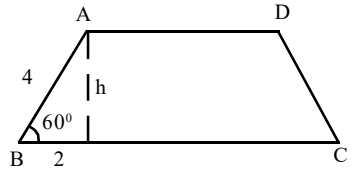
= $7x0.6428$

$= 4.5$



ii) Area = $\frac{1}{2}x12x4.5 = 27cm^2$

18.



$h = 2\sqrt{3}$

$AD = 6$

Area $= \frac{1}{2}(10+6)\,x\,2\sqrt{3}$

$= 8x2\sqrt{3} = 16\sqrt{3}\ cm^2$

19. i) 10 cm

ii) $10\sqrt{2}$

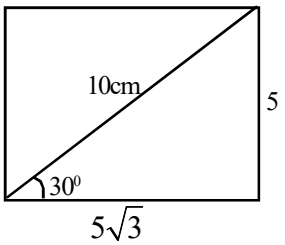
iii) Circumference of a circle $= 10\sqrt{2}\ \pi$

20. i) $\angle ACB = 30^0$

ii) Area of $\Delta BCD = \frac{1}{2}x5x5\sqrt{3} = \frac{1}{2}x25\sqrt{3}$

Area of quadrilateral ABCD $= 25\sqrt{3}\ cm^2$

21.



Area of rectangle $= 5x5\sqrt{3}$

$= 25\sqrt{3}\ cm^2$

22. i) $\angle ACB = 90^0$

ii) BC is $\frac{1}{2}$ of the diameter

AC is $\frac{\sqrt{3}}{2}$ of the diameter

iii) Area of $\Delta ABC = \frac{1}{2}x5x5\sqrt{3} = \frac{25}{2}\sqrt{3}$

23. Length of the chord = 2 x 4 x Sin40

= 8 x 0.64

= 5.12 cm

24. i) 4 cm

ii) Perimeter $= 4\sqrt{2}+8+4+4\sqrt{3} = 12+4\sqrt{2}+4\sqrt{3}$

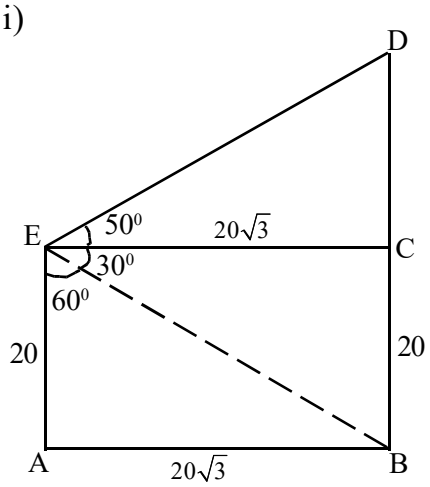
25. i) $\angle ADB = 65^0$

ii) Circum radius $= \frac{9}{2\,Sin\,65^0} = \frac{9}{2x0.91} = \frac{9}{1.82} = 4.95\ cm$

iii) BD = 2 x 4.95 x 0.42 = 4.15

Sides 9, 4.16, 9.9

26. i)



ii) $20\sqrt{3}$

iii) 40

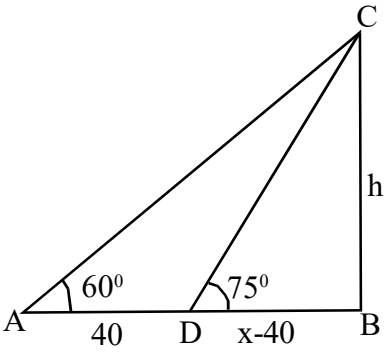
iv) $CD = 20\sqrt{3}\ x\ \tan 50 = 20x1.73x0.76$

= 26.29

Height of the building = BC + CD

= 20 + 26.29 = **46.29 m**

27. i)



.....x.....

ii) $\text{Tan}\,60 = \frac{h}{x}$

$\Rightarrow h = x\ x\tan 60$

$h = \sqrt{3}\ x$.....(1)

$\tan 75^0 = \frac{h}{x-40}$

$h = (x - 40).\tan 75$

$= (x - 40)x3.372$ .....(2)

From (1) & (2)

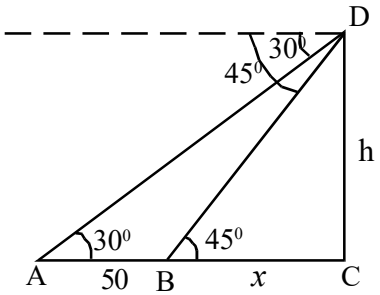
$\sqrt{3}x = (x - 40)x3.372$

$x = 82.24$

$h = \sqrt{3}$ x $82.24 = 1.732$ x 82.24

= 142.44 m

28. i)



ii) AB = 50m

$\tan 45 = \frac{h}{x}$

$\Rightarrow x = h$ .....(1)

$\tan\ 30^0 = \frac{h}{x+50} \Rightarrow \frac{1}{\sqrt{3}} = \frac{h}{x+50}$

$\Rightarrow \sqrt{3}xh = x + 50$ .....(2)

From (1) & (2)

$\sqrt{3}\ h = h + 50$

$h\left(\sqrt{3} - 1\right) = 50$

$h = \frac{50}{\sqrt{3}-1}$ $(\sqrt{3}+1)$

$= \frac{50}{\sqrt{3}-1}\ \lambda$ $(\sqrt{3}+1)$

$= \frac{50}{2}\left(\sqrt{3}+1\right) = 25x(1.73+1)$

= 68.3m

Chapter 7

COORDINATES

1. Which among the following is a point on x-axis?

   (1,1), (0, -3), (-3, 0), (3,2)

2. From the following find out the point on y-axis

   [(4,0), (5,0), (0,5), (0,0)]

3. Read the statement below

   1) (0,4) is a point on X-axis and (-2,0) is a point on y-axis

   2) y-coordinates of any point on X-axis is zero and x-coordinates of any point on y-axis is zero.

      i) Statement 1 is true, 2 is false

      ii) Statement 1 is false, 2 is true

      iii) Both the statements are correct, statement 2 is the correct reason for 1

      iv) Both the statements are correct, statement 2 is not the correct reason for 1

4. Read the following statements

   i) (5,0) is a point on X-axis

   ii) (4,0) is a point on y-axis

   iii) On a line parallel to X-axis, y - coordinates of all the points are same

   iv) On a line parallel to y-axis, x-coordinates of all the points are same

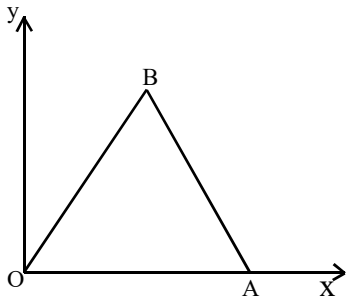
   A) i and ii are true

   B) i and iii are true

   C) ii and iii are true

   D) iii and iv are true

5. The triangle in the given figure is an equilateral triangle with perimeter 12cm



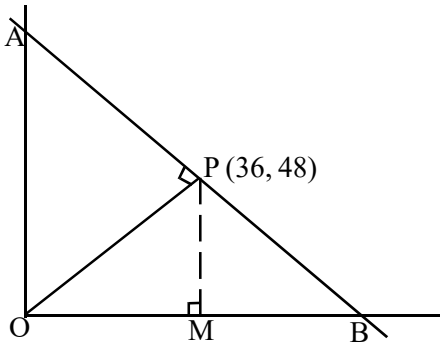
   i) Find the length of the sides of the triangle?
   ii) Find the coordinates of he vertex of the triangle?

6. (4,5) is a point on a circle with centre (2,2)
   i) Find the radius of the circle?
   ii) Is (5,0) is a point on the circle?
   iii) Find the coordinates of the point at where the circle cut the axes?

7. (8,11) is a point on the circle with centre at (2,3)
   i) Find the radius of the circle
   ii) Find out any four points on the circle

8. Check whether the points (5, -2), (6,4), (7,2) are the vertex of an isosceles triangle?

9. In the figure the point (36, 48) is the coordinate of the point P, find out the coordinates of A, B, M?



10. i) Find the coordinates of a point on X-axis, which is at a distance of 5 units away from the point (12,5)?
    ii) Find the coordinates of a point on X-axis, which is at a distance of 13 units away from the point (12,5)?

11. i) Draw X and Y axes mark points A(2,1), B(8,1) on the plane.
    ii) Draw an isosceles right angled triangle ABC with hypotenuse as AB?
    iii) Find the coordinates of 'C'?

Chapter 7

COORDINATES

ANSWERS

1. (-3, 0)

2. (0, 5)

3. ii

4. B

5. (i) 4 units

   ii) 0(0,0), A(4,0), B(2, $2\sqrt{3}$)

6. i) $\sqrt{13}$

   ii) Consider (2,2) and (0,5) Distance $=\sqrt{(5-2)^2+(0-2)^2}$

   iii) (5,0), (-1,0) $=\sqrt{9+4}=\sqrt{13}$, (0,5) is a point on the circle.

   (0,5), (0,-1)

7. i) 10

   ii) (12,3), (-8,3), (2,13), (2, -7)

8. i) A(5, -2), B(6,4), C(7,-2)

   AB = $\sqrt{37}$

   BC = $\sqrt{37}$

9. ΔOMP, $OP=\sqrt{48^2+36^2}$

   = 60

   $\therefore \Delta OPA$ and $\therefore \Delta OMP$ are similar

$$\frac{OA}{OP} = \frac{OP}{PM} \Rightarrow OA = \frac{OP^2}{PM}$$

$$= \frac{60^2}{48}$$

$$OA = 75$$

$\therefore \Delta OMP$ & ΔOPB are similar

$$\frac{OB}{OP} = \frac{OP}{OM} \Rightarrow OB = \frac{OP^2}{OM}$$

$$= \frac{60^2}{36}$$

$$OB = 100$$

$\therefore M(36, 0)$

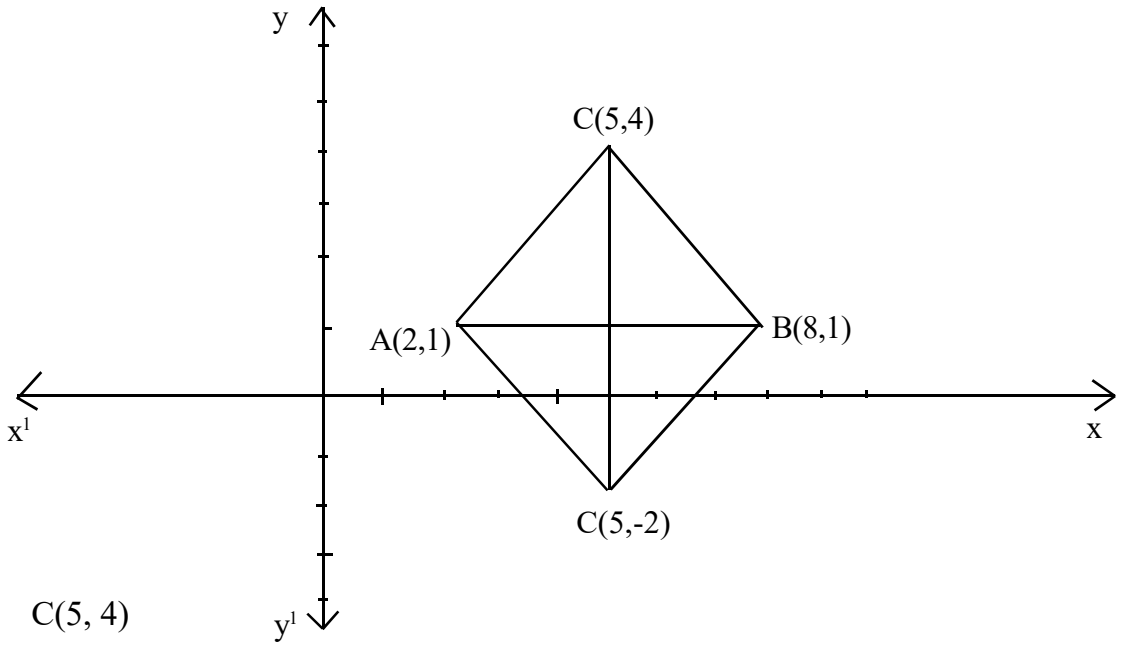
$A\ (0, 75)$

$B\ (100, 0)$

10. i) (12, 0)

ii) (0,0), (24,0)

11.



C(5, 4)

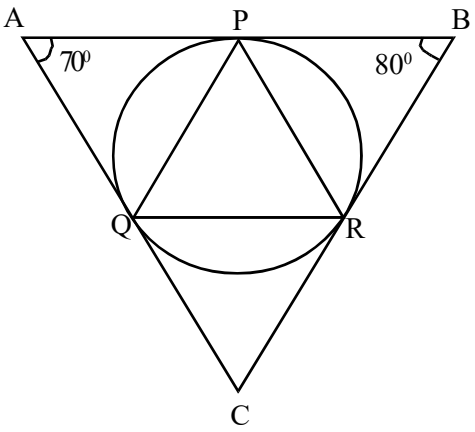
C(5, -2)

Chapter 8

TANGENTS

1. Calculate the length of the tangent from a point 15cm away from the centre of a circle of radius 9cm.

2. Draw a circle of radius 3 cm. Draw a triangle of angles 55^0, 63^0 with it's sides touching the circle.

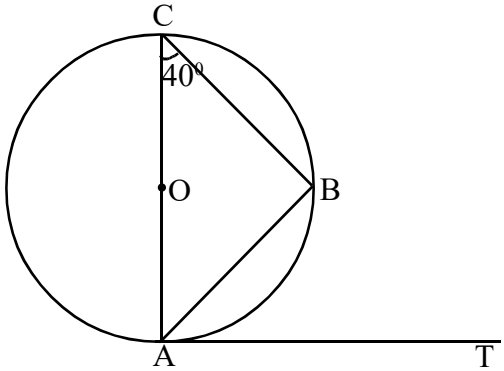
3. 

   Calculate the measures of all the angles of triangle PQR, where P, Q, R are the points touching the sides of the incircle of ΔABC.

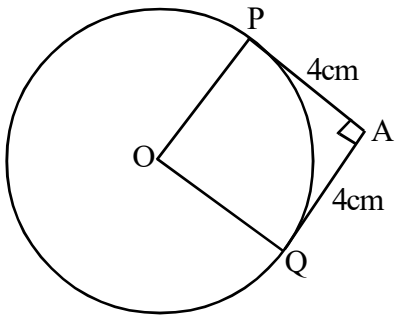
4. In the given figure AB is a chord of the circle and AOC is it's diameter such that $\angle ACB = 40^0$. If AT is the tangent to the circle at the point A, then $\angle BAT$ is equal to

   a) 60^0 b) 40^0

   c) 50^0 d) 30^0



5. In the figure the pair of tangents AP and AQ drawn from an external point to the circle with centre O, are perpendicular to each other and length of each tangent is 4cm. Then the radius of the circle is,



a) 10 cm b) 4 cm

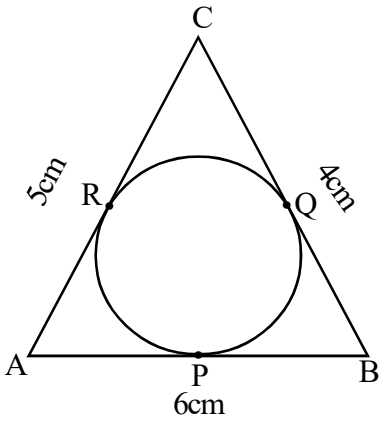
c) 7 cm d) 3 cm

6. The incircle of ΔABC touches it's sides at P, Q, R.

AB = 6cm

BC = 4cm

AC = 5cm

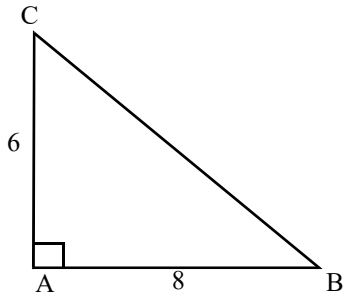


a) Denoting the length of AP by x.
Write the expression for the length of BQ and CQ.

b) Calculate the length of AP.

7. Calculate the radius of the incircle of a triangle with sides 3cm, 4cm, 5cm.

8. Find the radius of the incircle of the triangle given below.

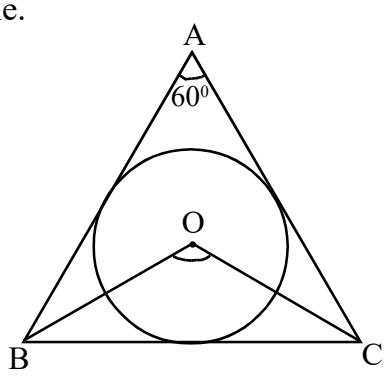


9. Calculate the angle based on the picture below.

O is the centre of incircle.

a) $\angle BOC$

b) $\angle B,\ \angle C$



10. Draw the triangle with sides 7cm, 8cm, 9cm and draw it's incircle.

Chapter 8

TANGENTS

ANSWERS

1. Length of the tangent = 12cm

2. Draw circle of radius 3cm. Mark 180 - 155, 180 - 63 on the centre of the circle. Draw tangents and complete the triangle.

3. The angles at the centre of the circle are 180 - 70, 180 - 80 and 180 - 30 then find 110^0, 100^0, 150^0.

   To find the angles of the triangle $\frac{110}{2} = 55^0$, $\frac{100}{2} = 50^0$, $\frac{150}{2} = 75^0$

4. b) $\angle BAT = 40^0$
5. b) 4 cm
6. a) AP = x, AR = x

   PB = 6-x, CR = 5-x

   BQ = 6-x, CQ = 5-x

   b) BQ + CQ = 4

   6-x + 5-x = 4

   11 - 2x = 4

   $\therefore$ 2x = 7

   $x = \frac{7}{2} = 3.5cm$

   is, AP = 3.5cm

7. Radius of incircle = $\sqrt{\frac{(S-a)(S-b)x(S-c)}{S}}$

   $S = \frac{a+b+c}{2} \quad = \frac{3+4+5}{2}$

   S = $\frac{12}{2} = 6$, $r = \sqrt{\frac{3x2x1}{6}}$ **= 1cm**

8. BC = 10 cm

$$r = \frac{(a+b-c)}{2} = \frac{6+8-10}{2}$$

$$= \frac{4}{2}$$

r = 2cm

9. a) $\angle BOC = 90 + \frac{60}{2} = 90 + 30 = 120^0$

b) $\angle B = 60^0 \quad \angle C = 60^0$

10. Construction

Chapter 9

POLYNOMIALS AND EQUATIONS

1. $103 \times 107 = 100^2 + 100 \times$ ..... $+ 21$

   A) 7 B) 3

   C) 10 D) 4

2. $75 \times$ ..... $= 70^2 + 8 \times 70 + 15$

   A) 78 B) 15

   C) 85 D) 73

3. $\left(x+\frac{1}{2}\right)\left(x-\frac{1}{2}\right) =$

   A) $x^2 - x - \frac{1}{4}$ B) $x^2 - \frac{1}{4}$

   C) $x^2 + \frac{1}{4}$ D) $x^2 - \frac{1}{4}x + \frac{1}{4}$

4. $(x-1) \times$ ..... $= x^2 + 5x - 6$

   A) $(x-3)$ B) $(x-2)$

   C) $(x+6)$ D) $(x-6)$

5. $(x+5)$ is a factor of the second degree polynomial

   A) $x^2 - 3x - 40$ B) $x^2 + 8x + 40$

   C) $x^2 + 8x - 40$ D) $x^2 + 3x - 40$

6. The second degree polynomial $x^2 + 3x - 10$ is the product of two first degree polynomials:

   A) $(x+2)(x+5)$ B) $(x-2)(x-5)$

   C) $(x+2)(x-5)$ D) $(x-2)(x+5)$

7. Statement 1 : If $x^2 + px + q = (x+2)(x+3)$ then p=5 and q=6

Statement 2 : The product of the first degree polynomials $(x+a)(x+b)$ is the second degree polynomial $x^2 + (a+b)x + ab$

Now choose the correct answer from those given below

A) Statement 1 is true and statement 2 is false

B) Statement 1 is false and statement 2 is true

C) Statement 1 and statement 2 are true and statement 2 is the reason of statement 1

D) Statement 1 and statement 2 are true and statement 2 is not the reason of statement 1

8. Statement 1 : A graph of the polynomial $x^2 - 7x + 12$ cross the x-axis at the points (4,0) and (3,0)

Statement 2 : The Second degree polynomial $(x+4)(x+3) = 0$ when $x = 4$ and $x = 3$

Now choose the correct answer from those given below

A) Statement 1 is true and statement 2 is false

B) Statement 1 is false and statement 2 is true

C) Statement 1 and statement 2 are true and statement 2 is the reason of statement 1

D) Statement 1 and statement 2 are true and statement 2 is not the reason of statement 1

9. Statement 1 : The second degree polynomial $x^2 + 1$ cannot be 0 for any value of x.

Statement 2 : x^2 will always a positive number for any value of x.

A) Statement 1 is true and statement 2 is false

B) Statement 1 is false and statement 2 is true

C) Statement 1 and statement 2 are true and statement 2 is the reason of statement 1

D) Statement 1 and statement 2 are true and statement 2 is not the reason of statement 1

10. Write the following second degree polynomials as the product of two first degree factors.

i) x^2-2 ii) x^2-9 iii) x^2-5 iv) x^2-1

11. Write the following second degree polynomials as the product of two first degree factors.

i) x^2-16x

ii) x^2+x

iii) x^2+8x

12. Write the following second degree polynomials as the product of two first degree polynomials.

i) x^2-x-42 ii) $x^2+7x+10$

iii) x^2-5x-6 iv) $x^2-12x+11$

13. In the second degree polynomial, $P(x)=x^2-2x-255$

i) Find P(1)?

ii) If $P(x)=0$, then find the value of x?

14. Write $x^2+23x+126$ as the product of two first degree polynomials.

15. If $(x+2)$ is the factor of $x^2+mx+18$

i) Find the value of m?

ii) Write the other factor of this polynomial?

16. $(x-4)$ is the factor of the polynomial x^2+7x-K

i) Find the value of K?

ii) Write the second degree polynomial as the product of two first degree polynomials?

17. At what points does the graph of the polynomial $4x^2-4x-3$ cross the x-axis?

18. At what points does the graph of the polynomial $3x^2 - 7x - 6$ cross the x -axis?

19. The graph of the polynomial $P(x) = x^2 - 5x + K$ cross the x-axis at the point (3,0)
    i) Find the value of K?
    ii) Find the other point at which the graph of this polynomial cross the x-axis?

20. Seven times a number subtracted from the square of this number gives 78. What is the number?

21. Five times a positive number subtract from twice its square gives 12. What is the number?

22. Product of two consecutive multiples of seven is 588. What are the number?

23. Sum of two number is 4 and thier product is 2. Find the numbers?

24. How many terms of the arithmetic sequence 7, 11, 15, ...starting from the first, are to be added to get 168?

25. From 2 onwards how many consecutive even numbers must be added to get the sum 552?

26. A rectangle is made by bending a 62 metre long rod. If the length of the diagonal is 25 metre, what are the length of the sides of this rectangle?

27. One side of a rectangle is 1 metre longer than the other side and diagonal is 9 metre longer than this shorter side. What are the length of the sides.

28. One side of a right triangle is one centimetre more than three times the length of the side perpendicular to it and the area of triangle is 77cm^2. Find the length of perpendicular side?

Chapter 9

POLYNOMIALS AND EQUATIONS

ANSWERS

1. C) 10

2. D) 73

3. B) $x^2 - \frac{1}{4}$

4. C) $(x+6)$

5. A) $x^2 - 3x - 40$

6. D) $(x-2)(x+5)$

7. C) Statement 1 and statement 2 are true and statement 2 is the reason of statement 1

8. A) Statement 1 is true and statement 2 is false

9. C) Statement 1 and statement 2 are true and statement 2 is the reason of statement 1

10. i) $\left(x+\sqrt{2}\right)\left(x-\sqrt{2}\right)$

    ii) $(x+3)\ (x-3)$

    iii) $\left(x+\sqrt{5}\right)\left(x-\sqrt{5}\right)$

    iv) $(x+1)\ (x-1)$

11. i) $x(x-16)$

    ii) $x(x+1)$

    iii) $x(x+8)$

12. i) $(x-7)\ (x+6)$ ii) $(x+5)\ (x+2)$

    iii) $(x-6)\ (x+1)$ iv) $(x-11)\ (x-1)$

13. i) -256

    ii) $x = 17,\ -15$

14. $(x+14)\ (x+9)$

15. i) $m = 11$

    ii) $(x+9)$

16. i) K = 44

    ii) $(x+11)\ (x-4)$

17. $\left(\frac{3}{2},0\right)\left(\frac{-1}{2},0\right)$

18. $(3,0)\ \left(\frac{-2}{3},0\right)$

19. i) K = 6

    ii) (2,0)

20. 13 or -6

21. 4

22. 21 and 28

23. $\left(2+\sqrt{2}\right)$ *and* $\left(2-\sqrt{2}\right)$

24. 8

25. 23

26. 24m, 7m

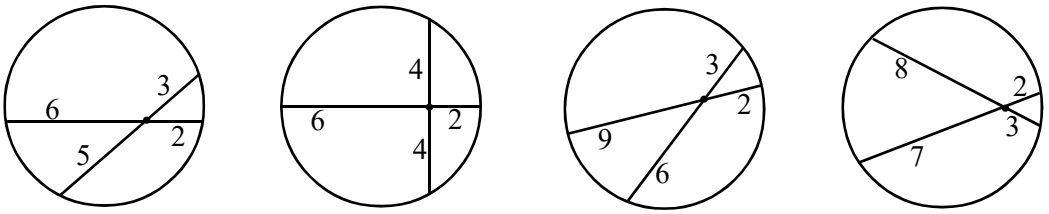
27. 20m, 21m

28. 7m, 22m

Chapter 10

CIRCLES AND LINES

1. In which figure all the lengths are marked with the correct measurement?



(a) (b) (c) (d)

2. Read the two statements given below.

Statement 1 : The length of tangents, drawn to two circles from any point on the line through intersection of these two circles, are equal.

Statement 2 : Whatever line we draw from a point outside a circle, intersecting the circle at two points, the product of the distances from the point to the points of intersection is the square of the length of the tangent from the point.

Choose the correct answer from those given below

a) Statement 1 is true and statement 2 is false

b) Statement 1 is false and statement 2 is true

c) Both the statements are true, statement 2 is the correct reason of statement 1

d) Both the statements are true, statement 2 is not the correct reason of statement 2

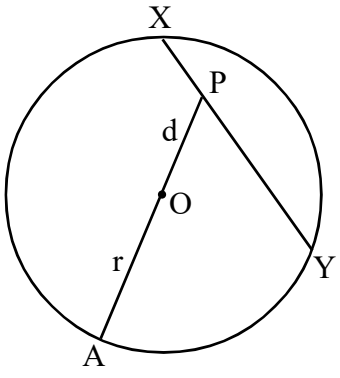
3. P is a point on the chord XY of the circle with centre O.

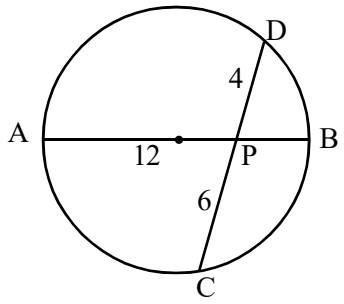
OA = r, OP = d then,

PX x PY = ...

a) r x d b) (r+d)xd

c) $(r+d)^2$ d) $r^2 - d^2$



4. 

Chord CD is drawn through point P on the diameter AB of the circle.

AP = 12cm, CP = 6cm, PD = 4cm

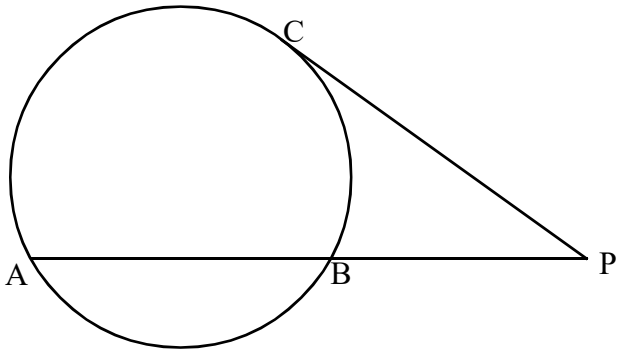
What is the radius of the circle?

(6cm, 7cm, 8cm, 9cm)

5. PC is tangent through C on the circle.

A, B are points on the circle such that $AB = BP = x$ then,

PC = ...

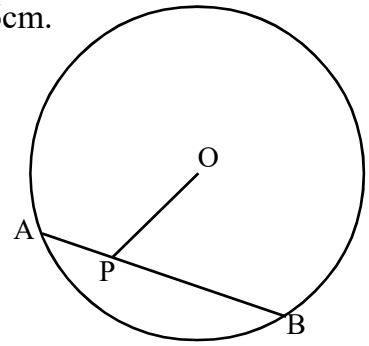


a) $2x$ b) x^2 c) $2x^2$ d) $\sqrt{2}\ x$

6. P is a point on the chord AB of circle with radius 6cm.

AP = 4cm, PB = 5cm.

Find OP?



7. Chords AB and CD intersect at a point P outside the circle.

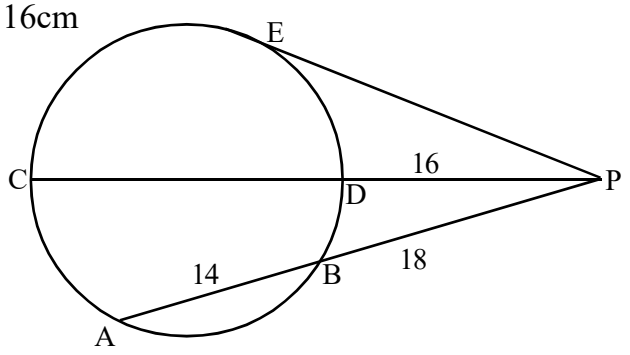
PE is tangent from P.

AB = 14cm, PB = 18cm, PD = 16cm

i) Find PD x PC?

ii) What is the length of chord CD?

iii) What is the length of tangent PE?



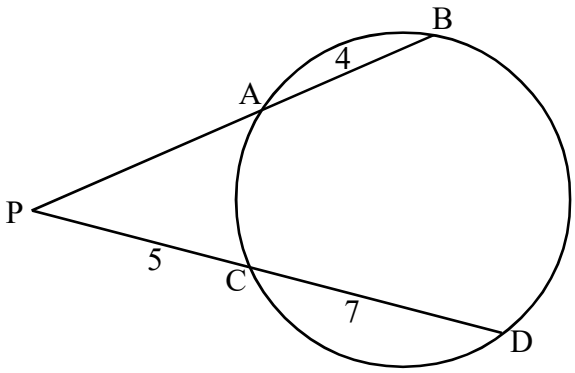
8. Chords AB and CD intersect at a point P outside the circle.

PC = 5cm, CD = 7cm, AB = 4cm.

i) Find PA x PB?

ii) If PA = x, What is the length of PB?

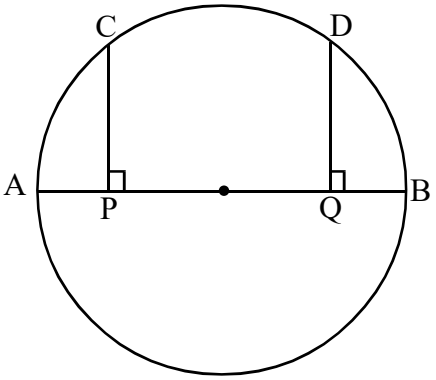
iii) Find the length of PA by forming a second degree equation in x.



9. Draw a rectangle of length 5cm and height 3cm. Draw a rectangle of side 6cm with same area.

10. Draw a square of area 20cm^2?

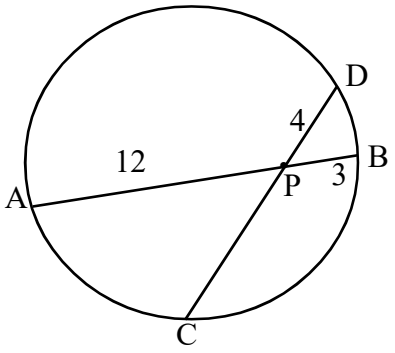
11.



In the figure PC, QD are drawn perpendicular to the diameter AB of the circle. If PC and QD are of same length, prove that AP and QB are of same length.

12. Chords AB and CD intersect at a point P inside the circle.

AP = 12cm, PB = 3cm, PD = 4cm.



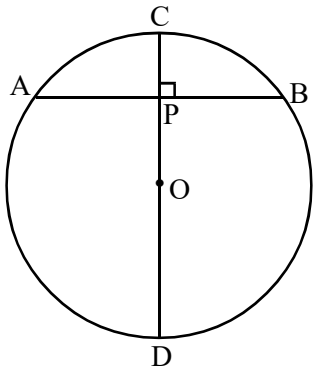
i) Find PC x PD?

ii) Find the length of PC?

iii) What is the length of chord EF with P as midpoint?

13. Draw an equilateral triangle of side $\sqrt{8}$ cm?

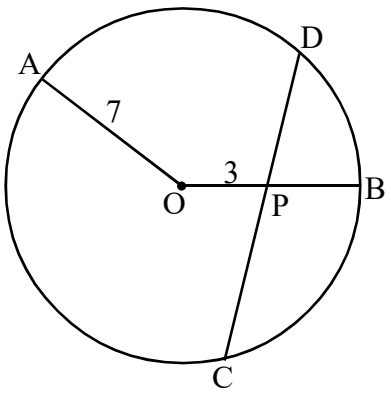
14. In the figure, chord AB is perpendicular to the diameter CD of the circle. AB = 16cm, CP = 4cm. Find the radius of the circle.



15. Draw a sqaure of area 25cm^2.
Draw a rectangle with one side 6cm and same area

16. Draw a rectangle of area 18cm^2.
Draw a square of equal area.

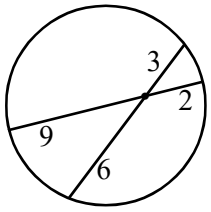
17. Chord CD pass through point P on radius OB of the circle with radius 7cm. OP = 3cm, CD = 13 cm. Find the length of the two parts of the chord CD?



Chapter 10

CIRCLES AND LINES

ANSWERS

1. 

(c)

2. c) Both the statements are true, statement 2 is the correct reason of statement 1

3. d) $r^2 - d^2$

4. 7cm

5. d) $\sqrt{2}\ x$

6. $AP \times PB = r^2 - d^2$

   $4 \times 5 = 6^2 - d^2 \Rightarrow d^2 = 16$

   $\Rightarrow d = 4cm$

7. i) $PD \times PC = PA \times PB = 32 \times 18 = 576$

   ii) $PD \times PC = 576$

   $16 \times PC = 576$

   $PC = \frac{576}{16} =$ **36**

   $CD = 36 - 16 =$ **20cm**

   iii) $PD \times PC = PE^2$

   $576 = PE^2 \Rightarrow PE =$ **24cm**

8. i) $PA \times PB = PC \times PD = 5 \times 12 = 60$

ii) $x + 4$

iii) $PA \times PB = PC \times PD$

$x(x+4) = 60$

$x^2 + 4x = 60$

$x^2 + 4x + 4 = 64$

$(x+2)^2 = 8^2$

$x + 2 = 8 \Rightarrow x = 6$

PA = 6cm

9. Construction

10. Construction

11. $AP \times PB = CP^2 = DQ^2 = AQ \times QB$

$AP(PQ + QB) = (AP + PQ)QB$

$AP \times PQ + AP \times QB = AP \times QB + PQ \times QB$

$AP \times PQ = PQ \times QB$

AP = QB

12. i) $PC \times PD = PA \times PB = 12 \times 3 = 36$

ii) $PC \times 4 = 36 \Rightarrow PC = \frac{36}{4} =$ **9cm**

iii) $EP \times PF = 36 \Rightarrow EP^2 = 36$ (EP = PF)

EP = 6cm

$\therefore$ EF = 6+6=12cm

13. Construction

14. $AP = PB = \frac{16}{2} = 8cm$

$AP \times PB = CP \times PD$

$64 = 4 \times PD$

$PD = \frac{64}{4} = 16$

$CD = 16 + 4 = 20$

$\therefore\ r = \frac{20}{2} =$ **10cm**

15. Construction

16. Construction

17. $CP \times PD = r^2 - d^2$

$= 7^2 - 3^2 = 40$

$CP + PD = 13$

$CP = x,\ PD = 13 - x$

$x(13 - x) = 40$

$13x - x^2 = 40$

$x^2 - 13x + 40 = 0$

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$= \frac{13 \pm \sqrt{13^2 - 4x40}}{2} = \frac{13 \pm 3}{2} = \frac{16}{2}$ *or* $\frac{10}{2}$ **= 8 or 5**

Chapter 11

GEOMETRY AND ALGEBRA

1. Read the following statements

   i) The diagonals of a parallelogram with opposite vertices (4,5) (1,3) cut each other at $(\frac{5}{2}, 4)$.

   ii) The diagonals of a parallelogram bisect each other

   a) Statement one is true, statement two is false

   b) Statement one is false, statement two is true

   c) Both the statements are true, two is the reason for one

   d) Both statements are true, two is not the correct reason for one

2. A circle is drawn with the line joining (2,3) and (6,5) as diameter, which among the following is the centre of the circle?

   a) (3,3) b) (4,4)

   c) (2,5) d) $(\frac{7}{2}, 4)$

3. Read the following statements

   i) In a parallelogram ABCD if A(1,3), B(3,4), D(2,5) then the coordinates of 'C' is (4,6)

   ii) In a parallelogram

   Sum of x-coordinates of opposite vertices are equal

   Sum of y - coordinates of opposite vertices are equal

   a) Statement one is true, statement two is false

   b) Statement one is false, statement two is true

   c) Both the statements are true, two is the reason for one

   d) Both statements are true, two is not the correct reason for one

4. Which among the following is a point on the line joining (2,5) & (6,7)?

   a) (4,6) b) (2,7)

   c) (6,5) d) (8,12)

5. Which of the following is the slope of the line joining (2,8), (4,9)?

a) 1 b) $\frac{1}{2}$

c) 2 d) $\frac{4}{7}$

6. If (-3,1) is a point on a line with slope $\frac{1}{3}$, which among the following is another point on this line?

a) (2, 0) b) (0, 2)

c) (1, 5) d) (5, 1)

7. Read the following statements

i) The midpoint of line joining (x_1, y_1) and (x_2, y_2) is $\left(\frac{x_1+x_2}{2}, \frac{y_1+y_2}{2}\right)$

ii) The point at which the line 3x-4y-12 = 0 cut x-axis is (3, 0).

iii) For any two points on a line, not parallel to either axis, the change in the y-coordinate is the change in the x-coordinate multiple by a fixed number.

a) i and ii are true

b) ii and iii are true

c) i and iii are correct

d) All are true

8. In a parallelogram ABCD, if A(1,3), B(3,4) and D(2,5)

a) Find the coordinates of 'C'?

b) Find the coordinates of midpoint the side 'AC'?

9. The midpoints of the sides of a triangle are (1,5), (4,1) and (4,5)

a) Find out the three vertices of the triangle?

b) Find the perimeter of the triangle?

10. The coordinates of the three vertices of a triangle are (3,5), (9,13), (10,6)

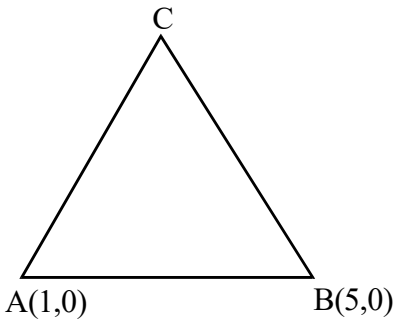
a) Prove that this triangle is isosceles?

b) Find the area of this triangle?

11. The coordinates of A and B are respectively (3,2), (8,7).
    a) Find the coordinates of a point 'P' such that AP : PB = 2:3?
    b) Find the coordinates of a point Q such that AQ : QB = 3:2?

12. In ΔABC the coordinates of A, B and C are A(-1,5), B(3,7), and C(1,1)
    a) Write the coordinates of midpoint of BC?
    b) Find the centroid of the ΔABC?

13. ΔABC is an equilateral triangle,
    the coordinates of A & B are
    (1,0) and (5,0) respectively
    a) Find the length of the side?
    b) Find the coordinates of the vectex 'C'?



14. P and Q are any two points on the line joining A(1,2) and B(7,5)
    Find out the coordinates of P and Q
    such that AP = PQ = QB

15. A(1,-3), B(3,3) are points on the same line
    a) Is the point (4,6) on the above line?
    b) Find out the coordinates of other two points on this line?

16. A(2,3) and B(6,5) are poins on the same line
    a) Find the slope of the line joining A&B?
    b) Find out the midpoints of AB?
    c) Write down the coordinates of any two points on this line?

17. The midpoint A (-1,1), B(2,7) are on the same line
    a) Write down the equation of the line?
    b) If (x,y) is on this line, prove that (x+1, y+2) is also on the same line?

18. (-3,1) is a point on a line with slope $\frac{1}{3}$
    a) Write down the equation of the line?

b) Is (3,4) a point on this line?

c) Find out any two points on this line?

19. In ΔOAB, if O(0,0), A(1,5), B(6,4),

a) Find the length of the side OA?

b) Prove that ΔOAB is a isosceles right angled triangle

c) Find the circum centre and radius?

20. A circle is drawn with diameter as the line joining (1,2) and (7,10)

a) Find the coordinates of the centre?

b) Find the radius of the circle?

c) Write the equation of the circle?

d) Find the coordinates of the points at which the circle cut Y-axis

21. If x+2y - 6=0 is the equation of a line,

a) Find the coordinates of the point that, cut the X-axis?

b) Find the coordinates of the point that the line cut the Y-axis?

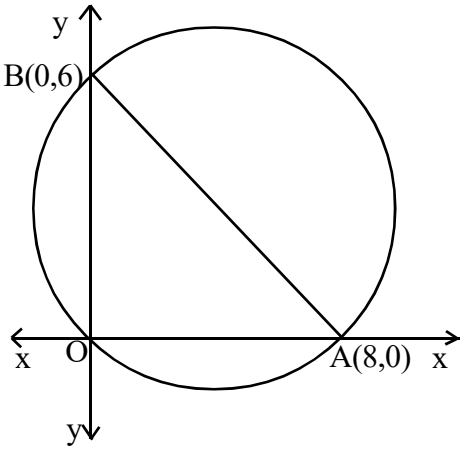
22. In the figure,

a) Find out the length of AB?

b) Find the coordinates of the centre of the circle?

c) Find the radius of the circle?

d) If (x, y) is a point on the circle
Prove $x^2 + y^2 - 8x - 6y = 0$



Chapter 11

GEOMETRY AND ALGEBRA

ANSWERS

1. c) Both the statements are true, two is the reason for one
2. b) (4,4)
3. c) Both the statements are true, two is the reason for one
4. a) (4,6)
5. b) $\frac{1}{2}$
6. b) (0, 2)
7. c) i and iii are true

8. Let C (x,y)

   a) $1+x = 3+2$

   $x = 5-1=4$

   $3+y = 4+5$

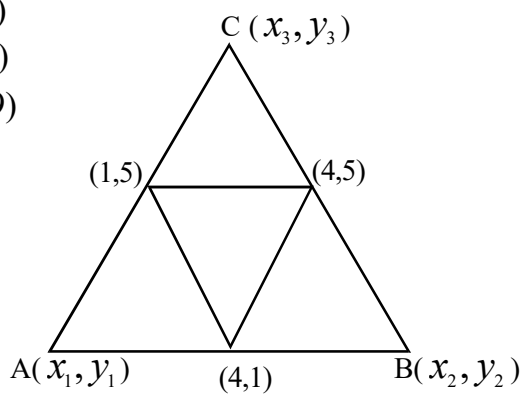
   $y = 9-3=6$

   C(4,6)

   b) Midpoint $= \left(\frac{1+4}{2}, \frac{3+6}{2}\right)$

   $= \left(\frac{5}{2}, \frac{9}{2}\right)$

9. a) A (1,1)
   B (7,1)
   C (1, 9)



b) $AB = \sqrt{(7-1)^2 + (1-1)^2} = \sqrt{6^2 + 0^2} = \sqrt{36} = 6$

$BC = \sqrt{(1-7)^2 + (9-1)^2} = \sqrt{(-6)^2 + 8^2}$

$= \sqrt{36+64} = \sqrt{100} = 10$

$CA = \sqrt{(1-1)^2 + (9-1)^2} = \sqrt{0^2 + 8^2} = \sqrt{64} = 8$

Perimeter = 6+10+8=24

10. a) A(3,5) B(9,13) C(10,6)

$AB = \sqrt{(9-3)^2 + (13-5)^2} = \sqrt{6^2 + 8^2} = \sqrt{100} = 10$

$BC = \sqrt{(10-9)^2 + (6-13)^2} = \sqrt{1^2 + (-7)^2} = \sqrt{1+49} = \sqrt{50}$

$CA = \sqrt{(10-3)^2 + (6-5)^2} = \sqrt{7^2 + 1^2} = \sqrt{49+1} = \sqrt{50}$

$BC = CA = \sqrt{50}$

ΔABC is an isoscless triangle

b) Area = $\frac{1}{2}bh$

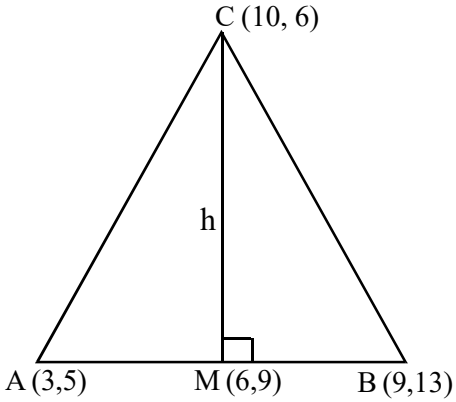
$= \frac{1}{2} x 10 x 5$

$= 5x5 = 25$

b = 10

$h = \sqrt{(10-6)^2 + (6-9)^2}$

$= \sqrt{4^2 + 3^2} \quad = \sqrt{16+9} = \sqrt{25} = 5$



11. a) If P(x,y)

$x = 3 + \frac{2}{5} x (8-3)$

$= 3 + \frac{2}{5} x 5 = 5$

$y = 2 + \frac{2}{5}(7-2) = 2 + \frac{2}{5} \text{ x } 5 = 4$

P(5, 4)

b) If Q (x,y)

$x = 3 + \frac{3}{5} \text{ x } 5 = 6$

$y = 2 + \frac{3}{5} \text{ x } 5 = 5$

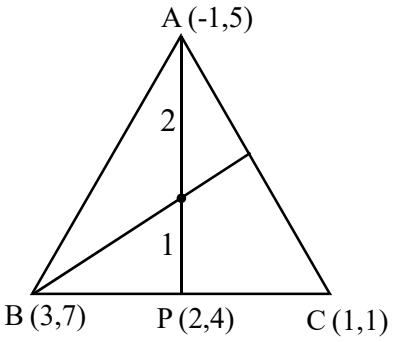
Q(6,5)

12. a) If P(x,y)

$x = \frac{3+1}{2} = \frac{4}{2} = 2$

$y = \frac{7+1}{2} = \frac{8}{2} = 4$

Mid point of BC is (2,4)



b) Centroid (x,y)

$x = {}^{-}1 + \frac{2}{3}(2 - {}^{-}1) = {}^{-}1 + \frac{2}{3} x 3 = {}^{-}1 + 2 = 1$

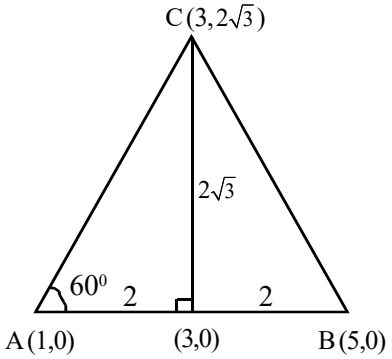
$y = 5 + \frac{2}{3}(4-5) = 5 + \frac{2}{3} x {}^{-}1 = 5 - \frac{2}{3} = \frac{13}{3}$

Centroid $= \left(1, \ {}^{13}/_{3}\right)$

13. Side $= \sqrt{(5-1)^2 + (0-0)^2}$

$= \sqrt{4^2} = \sqrt{16} = 4$

$C\left(3, \ 2\sqrt{3}\right)$



14. $AP = PQ = QB \Rightarrow 1:1:1$

If $P(x, y)$

$x = 1 + \frac{1}{3}(7-1)$

1+2 = 3

$y = 2 + \frac{1}{3}(5-2)$

= 2+1 = 3

If $Q(x, y)$

$x = 1 + \frac{2}{3}(7-1)$

1+4 = 5

$y = 2 + \frac{2}{3}(5-2)$

= 2+2=4

P(3,3), Q(5,4)

15. Slope = $\frac{y_1 - y_2}{x_1 - x_2} = \frac{3 - {}^{-}3}{3-1} = \frac{6}{2} = 3$

a) (3,3) (4,6)

$Slope = \frac{6-3}{4-3} = 3$

(4, 6) is a point or this line

b) y - difference = 3, x - difference = 1

(5, 9), (6, 12)

16. a) Slope = $\frac{5-3}{6-2} = \frac{2}{4} = \frac{1}{2}$

b) Midpoint $= \left(\frac{2+6}{2}, \frac{3+5}{2}\right) = (4,4)$

c) y - difference = 1

x - difference = 2

(6, 5), (8,6)

17. a) Slope $= \frac{7-1}{2-{}^{-}1} = \frac{6}{3} = 2$

$$\frac{y-1}{x-{}^{-}1} = 2$$

$$y-1 = 2(x+1)$$

$$2x-y+3 = 0$$

b) $x = (x+1)$

$y = (y+2)$ then,

$2(x+1)-(y+2)+3 = 2x+2-y-2+3$

$\Rightarrow 2x-y+3 = 0$

$(x+1,\ y+2)$ lies on this line

Or

Slope $= \frac{2}{1}$

Change in $x = 1$

Change in y $= 2$

$\therefore$ If (x, y) is a point on this line,

$(x+1,\ y+2)$ is also on this line

18. a) Slope $\frac{1}{3}$

$$\frac{(y-1)}{x-{}^{-}3} = \frac{1}{3}$$

$$3(y-1) = x+3$$

$$x-3y+6 = 0$$

b) x=3, y=4 then,

$3-3x4+6 = 3-12+6$

$= 9-12 = {}^{-}3 \neq 0$

(3,4) not in this line

c) y - difference = 1

x - difference = 3

(-3,1) is a point

(-3+3, 1+1) = (0,2)

(0+3, 2+1) = (3,3) is a point on this line

19. a) $OA = \sqrt{1^1 + 5^2} = \sqrt{1+25} = \sqrt{26}$

b) $AB = \sqrt{(6-1)^2 + (4-5)^2} = \sqrt{25+1} = \sqrt{26}$

$OA = AB = \sqrt{26}$

$OB = \sqrt{6^2 + 4^2} = \sqrt{36+16} = \sqrt{52}$

$(OA)^2 + (AB)^2 = (OB)^2$

ΔOAB is isosceles right angled triangle

c) Circum centre is the Midpoint of the hypotenuse

$$= \left(\frac{6+0}{2}, \frac{4+0}{2}\right) = (3,2)$$

$$\text{radius} = \frac{\sqrt{52}}{2} = \frac{2\sqrt{13}}{2} = \sqrt{13}$$

20. a) Centre $= \left(\frac{7+1}{2}, \frac{10+2}{2}\right) = (4,6)$

b) Radius $= \sqrt{(4-1)^2 + (6-2)^2}$

$= \sqrt{3^2 + 4^2} = \sqrt{9+16} = \sqrt{25} = 5$

c) $(x-4)^2 + (y-6)^2 = 5^2$

$x^2 - 8x + 16 + y^2 - 12y + 36 = 25$

$x^2 + y^2 - 8x - 12y + 27 = 0$

d) $x = 0$ then, $y^2-12y+27 = 0$

$(y-9)(y-3) = 0$

$y = 9, \quad y = 3$

Points are (0,9), (0,3)

21. a) $y = 0$

$x+0-6=0$

$x=6$

Point (6,0)

b) $x = 0$

$0 + 2y - 6 = 0$

$2y = 6$

y = 3

22. a) $AB=\sqrt{(8-0)^2+(0-6)^2}$

$= \sqrt{64+36} = \sqrt{100} = 10$

b) Centre $= \left(\frac{8+0}{2}, \frac{6+0}{2}\right)$

$= (4,3)$

c) Radius $= \frac{AB}{2} = \frac{10}{2} = 5$

d) $(x-4)^2+(y-3)^2 = 5^2$

$x^2-8x+16+y^2-6y+9=25$

$x^2+y^2-8x-6y=0$

Chapter 12

SOLIDS

MCQ

1. The lateral faces of a square pyramid are equilateral triangles. If the length of one base edge is 20 centimetres. What will be the measure of its slant height in cm.

   a) 10 b) $10\sqrt{2}$

   c) $10\sqrt{3}$ d) 20

2. Two square pyramids have the same volume. The base edge of one is half that of the other. How many times is the height of the second pyramid the height of the first?

   a) 2 b) 3

   c) 4 d) 8

3. From the circle of radius 12centimetres a sector of central angle 90° is cut out and made in to a cone. What is the base radius of the cone.

   a) 3 b) 6

   c) 9 d) 12

4. What is the slant height of a cone made by rolling up a sector of central angle 60° cut out from a circle of radius 10 centimetres?

   a) 2.5 b) 5

   c) 10 d) 12

5. What is the ratio of the base-radius and slant height of a cone made by rolling up a semicircle?

   a) 1:1 b) 1:2

   c) 2:1 d) 1:4

6. The volumes of two spheres are in the ratio 125:64. What is the ratio of the radius?

   a) 5 : 3 b) 3 : 5

   c) 5 : 4 d) 4 : 5

7. Statement 1 : If the height of the Cone is 24 centimetres and diameter of the base is 14 cm, then the slant height of the Cone is 15 centimetres.

Statement 2 : If 'r' the radius and 'h' is the height of the Cone, the slant height $\ell = \sqrt{n^2 + r^2}$

a) Statement 1 is true, Statement 2 is false

b) Statement 1 is false, Statement 2 is correct

c) Both the statements are true, Statement 2 is the reason for Statement 1

d) Both the statements are true, Statement 2 is not a reason for Statement 1

8. Statement 1 : Area of the curved surface of a cone is half the product of the base circumference and the slant height

Statement 2 : The area of the curved surface of a cone is the area of the sector used to make it.

a) Statement 1 is true, Statement 2 is false

b) Statement 2 is true, Statement 1 is false

c) Both answers are true, Statement 2 is explanation of Statement 1

d) Both answers are true, Statement 2 is not the explanation of Statement 1

9. Statement 1 : To make a conical hat of base radius 8cm and stant height 30cm. We need a paper of 240π square centimetres.

Statement 2 : The area of the curved surface of a cone is half the product of the base circumferance and slant height.

a) Statement 1 is true, Statement 2 is false

b) Statement 1 is false, Statement 2 is correct

c) Both the statements are true, Statement 2 is the reason for Statement 1

d) Both the statements are true, Statement 2 is not a reason for Statement 1

LONG ANSWER QUESTIONS

1. A square of side 6 centimetres, and four isosceles triangles of base 6 centimetres and height 7 centimetres, are to to be put together to make a square pyramid. How many square centimetres of paper are needed?

2. A toy is in the shape of a square pyramid of base edge 12 centimetres and slant height 10 centimetres. What is the total cost of painting 500 such toys, at 80 rupees per square metre?

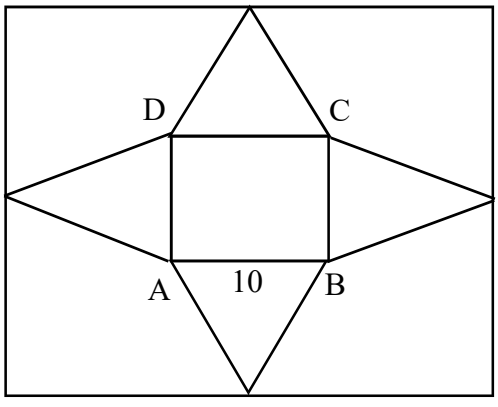
3. The lateral faces of a square pyramid are equilateral triangles and the length of a base edge is 20 centimetres. What is its surface area?

4. Prove that in any square pyramid, the squares of the height, slant height and lateral edge are in arithmetic sequence.

5. A square pyramid of base edge 10 centimetres and height 12 centimetres is to be made of paper. What should be the dimensions of the triangles?

6. A vessel (without lid) in the shape of a square pyramid, made from a metallic sheet is of base perimeter 80 centimetres and slant height 26 centimetres.
   i) How many square centimetres of the metallic sheet was needed to make the vessel?
   ii) Calculate the height of the vessel.
   iii) What is the capacity of the vessel in litres?

7. The figure of a square sheet of paper is shown below. Length of one side of the paper sheet is 36cm and AB = 10centimetres. The shaded portion is cut out and folded into a square pyramid



i) What is the slant height of the pyramid?

ii) What is the height of the pyramid?

iii) Find the volume of the pyramid.

8. The length of base edge of a square pyramid is 12centimetres and slant height is 10centimetres.

   i) What is the height of the pyramid?

   ii) calculate the volume of the pyramid.

9. The edges of a square pyramid are of equal length. The height is $6\sqrt{2}$ centimetres.

   i) Find the length of the base edge

   ii) Find the volume of the pyramid.

10. The base edges of two square pyramids are in the ratio 2 : 3 and their heights in the ratio 1 : 2. The volume of the first is 180 cubic centimetres. What is the volume of the second?

11. All edges of a square pyramid are 18 centimetres. What is its volume?

12. The slant height of a square pyramid is 25 centimetres and its surface area is 896 square centimetres. What is its volume?

13. What is the surface area of a square pyramid of base perimeter 64 centimetres and volume 1280 cubic centimetres?

14. A cone of radius 12 centimetres is to be made by folding a sector cut from a circle of radius 20 centimetres.

   i) What should be the central angle of the sector?

   ii) Calculate the curved surface area of the cone.

15. From a circle of radius 15 centimetres, a sector of central angle 120° is cut out and rolled up to make a cone.

   i) What is the slant height of the cone?

ii) What is the base radius of the cone?

iii) Calculate the curved surface area of the cone.

16. A cone of height 8cm is to be made using a sector of circle of radius 10 centimetres.
    i) What should the central angle of the sector.
    ii) calculate the volume and curved surface area of the cone.

17. A cone is made from sector of radius 10cm and central angle 216°
    i) What is the height and radius of the cone.
    ii) Find the volume of the cone

18. The base radii of two cones are in the ratio 3 : 5 and their heights are in the ratio 2 : 3. What is the ratio of their volumes?

19. Two cones have the same volume and their base radii are in the ratio 3 : 5. What is the ratio of their heights?

20. What is the area of the curved surface of a cone of base radius 12 centimetres and slant height 25 centimetres?

21. What is the surface area of a cone of base diameter 30 centimetres and height 40 centimetres?

22. A conical fire work is of base diameter 24 centimetres and height 5 centimetres. 5000 such fireworks are to be wrapped in colour paper. The price of the colour paper is 3 rupees per square metre. What is the total cost?

23. Prove that for a cone made by rolling up a semicircle, the area of the curved surface is twice the base area.

24. The base radius and height of a solid metal cylinder are 12 centimetres and 20 centimetres. By melting it and recasting, how many cones of base radius 4 centimetres and height 5 centimetres can be made?

25. Diameters of two hemispheres are in the ratio 4: 3.
    i) Write the ratio of their radii.
    ii) Find the ratio of their surface areas.
    iii) If the surface area of the first hemisphere is 144 square centimetres, what is the surface area of the other?

26. A solid metal hemisphere of radius 10 centimetres is melted and recast into small solid spheres of radius 1 centimetre each. How many such spheres can be made?

27. The sides of rectangular prism mare 4, 6, 10 centimetres.
    i) What is the maximum volume of a square pyramid which can be carved out from the prism?
    ii) Find the maximum volume of a sphere which can be curved out from the prism.

28. The radius of a solid metal sphere is 6 centimetres.
    i) Find the volume of the sphere.
    ii) This sphere is melted and recast in to a solid cone of radius 6cm. Find the height of the cone

29. i) What is the radius of the largest sphere that can be curved from a cube of edge 12 centimetres?
    ii) Find the surface area and volume of the sphere?
    iii) What is the volume of the cone of maximum size that can be curved from a cube of edge 12 centimetres?

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Chapter 12

SOLIDS

ANSWERS

MCQ

1. c) $10\sqrt{3}$
2. c) 4
3. a) 3
4. c) 10
5. b) 1:2
6. c) 5 : 4
7. B) Statement 1 is false, Statement 2 is correct
8. c) Both answers are true, Statement 2 is explanation of Statement 1
9. c) Both the statements are true, Statement 2 is the reason for Statement 1

LONG ANSWERS

1. a = 6cm $\quad \ell = 7cm$

   Total Surface Area = $a^2 + 2a\ell$

   = 6^2+2 x 6 x 7

   = 36 + 84

   = 120 cm^2

2. a = 12 cm $\quad \ell = 10cm$

   Total Surface Area = $a^2 + 2a\ell$

   = 12^2 + 2 x 12 x 10

   = 144 + 240

$= 384 \text{ cm}^2$

Total area = 384 x 500

$= 192000 \text{ cm}^2$

$$\text{Total Cost} = \frac{192000}{10000} x 80$$

= Rs. 1536

3. a = e = 20cm

$$e^2 = \ell^2 + \left(a/2\right)^2$$

$$\ell^2 = 20^2 - 10^2$$

$$\ell^2 = 300$$

$$\ell = \sqrt{300} = 10\sqrt{3}$$

$$\text{Total Surface Area} = a^2 + 2a\ell$$

$$= 20^2 + 2x20x10\sqrt{3}$$

$$= 400 + 400\sqrt{3}$$

$$= 400 + 400x1.73$$

$$= 400 + 692$$

= 1092 cm²

4. $h^2,\ \ell^2,\ e^2$

$$\ell^2 - h^2 = \left(a/2\right)^2$$

$$e^2 - \ell^2 = \left(a/2\right)^2$$

$\therefore$ The common difference is $\left(a/2\right)^2$

$\therefore$ $h^2,\ \ell^2,\ e^2$ in arithmetic sequence

5. a = 10 cm, h = 12cm

$\ell^2 = h^2 + \left(a/2\right)^2$

$= 12^2 + 5^2 = 144 + 25 = 169$

$= \ell = 13cm$

$e^2 = \ell^2 + \left(a/2\right)^2 = 13^2 + 5^2$

$= 169 + 25 \qquad e^2 = 194$

e = $\sqrt{194}$

6. 4a = 80 cm

$a = \frac{80}{4} = 20cm$

$\ell = 26cm$

i) Total Surface Area = $a^2 + 2a\ell$

$= 20^2 + 2 \times 20 \times 26$

$= 400 + 1040$

= 1440 cm²

ii) $\ell^2 = h^2 + \left(a/2\right)^2$

$h^2 = \ell^2 - \left(a/2\right)^2$

$= 26^2 - 10^2$

$= 676 - 100 = 576 = 24^2$

h = 24 cm

iii) $V = \frac{1}{3}a^2h$

$= \frac{1}{3} x 20^2 x 24 \qquad = 400x8 =$ **3200 cm³**

$= \frac{3200}{1000}\ell = \mathbf{3.2\ell}$

7. i) a = 10 cm

ii) $a + 2\ell = 36$

$2\ell = 36 - 10 = 26$

$\ell = 13\ cm$

$\ell^2 = h^2 + \left(a/2\right)^2$

$h^2 = \ell^2 - \left(a/2\right)^2$

$= 13^2 - 5^2$

$= h^2 = 12^2$

$h = 12cm$

iii) Volume $= \frac{1}{3}a^2h$

$= \frac{1}{3}10^2 x12$

$= \frac{1200}{3}\ cm^3$

= 400 cm^3

8. a = 12 cm $\ell = 10cm$

$\ell^2 = h^2 + \left(a/2\right)^2$

$h^2 = 10^2 - 6^2$

$= 100 - 36$

$h^2 = 64 = 8^2$

h = 8cm

Volume $= \frac{1}{3}a^2h$

$= \frac{1}{3}x12^2 x8$

= 384 cm^3

9. a = e, $h = 6\sqrt{2}\ cm$

$$e^2 = \ell^2 + \left(a/2\right)^2$$

$$a^2 = \ell^2 + \frac{a^2}{4}$$

$$\ell^2 = \frac{a^2}{1} - \frac{a^2}{4} = \frac{3a^2}{4}$$

$$\ell^2 = h^2 + \left(a/2\right)^2$$

$$\frac{3a^2}{4} = h^2 + \frac{a^2}{4}$$

$$h^2 = \frac{2a^2}{4} = \frac{a^2}{2}$$

$a^2 = h^2 \text{ x } 2$

$$= \left(6\sqrt{2}\right)^2 x2 = 36x2x2$$

$a^2 = 72 \text{ x } 2 = 144$

a = 12 cm

$$V = \frac{1}{3}a^2h$$

$$= \frac{1}{3}x12^2x6\sqrt{2}$$

$$= 144x2\sqrt{2}$$

$$= 288\sqrt{2}\ \text{cm}^3$$

10. $a_1 : a_2 = 2: 3$

$a_1 = 2a \quad a_2 = 3a$

$h_1 : h_2 = 1 : 2$

$h_1 = 1h \quad h_2 = 2h$

$$\frac{V_1}{V_2} = \frac{\frac{1}{3}a_1^2h_1}{\frac{1}{3}a_2^2h_2}$$

$$= \frac{\frac{1}{3}(2a)^2 x1h}{\frac{1}{3}(3a)^2 x2h}$$

$$= \frac{4a^2 x1h}{9a^2 x2h}$$

$$\frac{V_1}{V_2} = \frac{2}{9}$$

$$2\ V_2 = 9V_1$$

$$V_1 = 240\ cm^3$$

$$2\ V_2 = 9x240$$

$$V_2 = \frac{9x240}{2} \quad = 9x\ 120 \quad \textbf{= 1080 cm}^3$$

11. $a = e = 18$ cm

$$e^2 = \ell^2 + \left(a/2\right)^2$$

$$\ell^2 = e^2 - \left(a/2\right)^2$$

$$\ell^2 = 18^2 - 9^2 = 324 - 81$$

$$\ell^2 \quad \textbf{= 243}$$

$$h^2 = \ell^2 - \left(a/2\right)^2$$

$$h^2 = 243 - 9^2$$

$$= 243 - 81 = 162$$

$$h = \sqrt{162} = \sqrt{81x2}$$

$$h = 9\sqrt{2}$$

$$V = \frac{1}{3}a^2 h$$

$$= \frac{1}{3}18^2 x9\sqrt{2}$$

$= 324x3\sqrt{2}$

= 972 x 1.41

= 1370.52 cm^3

12. $\ell = 25$cm

Surface area = $a^2 + 2a\ell = 896$

$a^2 + 2xax25 = 896 \quad \Rightarrow a^2 + 50a = 896$

$a^2+50a + 25^2 = 896 + 25^2$

$(a+25)^2 = 1521 = 39^2$

a+25 = 39

a = 39 - 25 = **14**

$\ell^2 = h^2 + \left(a/2\right)^2$

$h^2 = 25^2 - 7^2$

= 625 - 49

= 576 = 24^2

h = 24

V = $\frac{1}{3} a^2 h$

= $\frac{1}{3} x14^2 x24$

= 196 x 8

= 1568 cm^3

13. 4a = 64cm

$a = \frac{64}{4} = 16cm$

$V = \frac{1}{3} a^2 h$

$= \frac{1}{3} x16^2 xh = 1280$

$h = \frac{1280x3}{256}$

h = 15cm

$$\ell^2 = h^2 + \left(a/2\right)^2$$

$$\ell^2 = 15^2 + 8^2$$

$$\ell^2 = 225 + 64$$

$$\ell^2 = 289$$

$$\ell = 17cm$$

Total Surface Area $= a^2 + 2a\ell$

$= 16^2 + 2x16x17$

= 800 cm²

14. R = 20cm

r = 12 cm

i) $\frac{x}{360} = \frac{r}{R}$

$\frac{x}{360} = \frac{12}{20}$

$x = \frac{12}{20} x360$

= 216⁰

ii) $\ell = R = 20$

Curved Surface Area $= \pi r\ell$

$= \pi x12x20$

$= 240\pi\ cm^2$

15. R = 15 cm x = 120⁰

i) $R = \ell = 15\ cm$

ii) $\frac{120}{360} = \frac{r}{15}$

$r = \frac{120x15}{360} = 5\ cm$

Curved Surface Area = πrl

$= \pi x 5 x 15$

$= 75\pi \ cm^2$

16.i) h = 8 cm

R = 10cm $R = \ell = 10cm$

$\ell^2 = h^2 + r^2$

$r^2 = \ell^2 - h^2$

$r^2 = 10^2 - 8^2 = 6^2$

r = 6cm

$\frac{x}{360} = \frac{r}{R}$

$\frac{x}{360} = \frac{6}{10}$

$x = 36x6 = 216^0$

ii) Curved Surface Area $= \pi r\ell$

π x 6 x 10

$60\pi \ cm^2$

Volume $= \frac{1}{3}\pi r^2 h$

$= \frac{1}{3}\pi \ 6^2 x8 = 96\pi cm^3$

17. R = 10 cm x = 216^0

i) $\frac{x}{360} = \frac{r}{R}$

$\frac{216}{360} = \frac{r}{10}$

$r = \frac{10x216}{360} = 6 \ cm$

ii) Volume = $\frac{1}{3}\pi r^2 h$

$\frac{1}{3}\pi x6^2 x8 = 96\pi \ cm^3$

$\ell^2 = h^2 + r^2$

$h^2 = 10^2 - 6^2$

$h^2 = 8^2$

h = 8 cm

18. $r_1 : r_2 = 3:5$

$r_1 = 3r, \quad r_2 = 5r$

$h_1 : h_2 = 2 : 3$

$h_1 = 2h, \quad h_2 = 3h$

$$\frac{V_1}{V_2} = \frac{\frac{1}{3}\pi r_1^2 h_1}{\frac{1}{3}\pi r_2^2 h_2}$$

$$= \frac{(3r)^2 (2h)}{(5r)^2 (3h)}$$

$$= \frac{9x2}{25x3} = \frac{6}{25}$$

$V_1 : V_2 = 6 : 25$

19. $V_1 = V_2$

$$\frac{1}{3}\pi r_1^2 h_1 = \frac{1}{3}\pi r_2^2 h_2 \quad \Big| \quad r_1 : r_2 = 3:5$$

$$(3r)^2 h_1 = (5r)^2 h_2 \quad \Big| \quad r_1 = 3r, \; r_2 = 5r$$

$$\frac{h_1}{h_2} = \frac{25}{9}$$

$h_1 : h_2 = 25 : 9$

20. $r = 10cm \quad \ell = 25\ cm$

Curved Surface Area $= \pi rl$

$= \pi x10x25$

$= = 250\pi\ cm^2$

21. $r = \frac{30}{2} = 15cm \quad h = 40cm$

$\ell^2 = r^2 + h^2 = 15^2 + 40^2$

$= 225 + 1600$

$= 1825$

$\ell = \sqrt{25x73}$

$= 5\sqrt{73}$

Curved Surface Area $= \pi r\ell$

$= \pi x15x5\sqrt{73}cm^2$

$= 75\sqrt{73}\pi cm^2$

Total Surface area = $\pi r^2 + \pi rl = \pi x15^2 + 75\sqrt{73}\ \pi$

$225\pi + 75\sqrt{73}\pi\ cm^2$

22. r = 12 cm h = 5cm

$\ell^2 = h^2 + r^2 = 5^2+12^2 = 13^2$

$\ell = 13cm$

Total Surface Area $= \pi r^2 + \pi rl$

$= \pi x12^2 + \pi x12x13$

$= 144\pi + 156\pi = 300\pi\ cm^2$

Total area 5000 cones

$= 300\pi x5000cm^2$

$= \frac{300\pi x5000}{10000} m^2 = 30\pi x5 = 150\pi\ m^2$

Total Cost $= 150x3.14x3$

= 1413

23. x = 180^0

$\frac{x}{360} = \frac{r}{R}$

$\frac{180}{360} = \frac{r}{R} \Rightarrow \frac{1}{2} = \frac{r}{R}$

R = 2r $R = \ell$

Curved Surface Area $= \pi r\ell$

$= \pi x r x 2r$

$= 2\pi r^2$

Curved Surface Area = 2 x Base Area

24. r = 12cm h = 20cm

Volume of Cylinder = $\pi r^2 h$

$= \pi x 12^2 x 20$

Volume small cone $= \frac{1}{3}\pi x 4^2 x 5$

No. of cones = $\frac{\text{Volume of Cylinder}}{\text{Volume of one cone}}$

$= \frac{\pi x 12 x 12 x 20}{\frac{1}{3}\pi x 4 x 4 x 5}$ = 3x3x3x4 = **108**

25. a) $d_1 : d_2 = 4 : 3$

$r_1 : r_2 = \frac{d_1}{2} : \frac{d_2}{2} = 4 : 3$

ratio of surface area hemispheres

$S_1 : S_2 = 3\pi r_1^2 = 3\pi r_2^2$

$= r_1^2 : r_2^2 = \left(\frac{4}{3}\right)^2 = 16 : 9$

$S_1 : S_2 = 16 : 9$

$\frac{S_1}{S_2} = \frac{16}{9}$

$\Rightarrow 9S_1 = 16S_2$

$9x144 = 16S_2$

$\frac{9x144}{16} = S_2$

S_2 = 81 cm²

26. r = 10 cm

Volume of hemisphere = $\frac{2}{3}\pi x10^3$

Volume of one small sphere = $\frac{4}{3}\pi x1^3$

No.of spheres = $\frac{\frac{2}{3}\pi x10^3}{\frac{4}{3}\pi x1^3}$

$= \frac{10^3}{2} = 500$

27. i) Maximum Volume of square pyramid = $= \frac{1}{3}a^2h$

$= \frac{1}{3} 4^2 x10$

$= \frac{160}{3} = 53.33cm^3$

ii) **Diameter = 4cm r = 2 cm**

Maximum volume of sphere = $\frac{4}{3}\pi r^3$

$= \frac{4}{3}\pi x2^3 = \frac{32}{3}\pi cm^3$

28. r = 6cm

$V = \frac{4}{3}\pi r^3$

$= \frac{4}{3}\pi x6^3$

$= 288\pi cm^3$

Volume of cone = $\frac{1}{3}\pi r^2h$

$= \frac{1}{3}\pi x6^2 xh$

$\Rightarrow \frac{1}{3}\pi x6^2 xh = \frac{4}{3}\pi x6^3$

$$h = \frac{\frac{4}{3}\pi x6x6x6}{\frac{1}{3}\pi x6x6}$$

h = 24cm

29. i) r = 6cm

Total Surface Area $= 4\pi r^2$

$= 4\pi x6^2$

$= 144\pi\ cm^2$

ii) Volume sphere $= \frac{4}{3}\pi r^3$

$= \frac{4}{3}\pi x6^3$

$= 288\pi\ cm^3$

iii) $r = \frac{12}{2} = 6cm$

h = 12cm

Volume $= \frac{1}{3}\pi r^2 h$

$= \frac{1}{3}\pi x6^2 x12$

$= 144\pi\ cm^3$

Chapter 13

STATISTICS

1. In an Examination, scores obtained by 11 students are given below.

   15, 35, 20, 18, 40, 32, 28, 50, 45, 27, 31

   a) Find the mean score?

   b) Find the Median score?

2. Marks obtained by some students are given below.

   Find the Median mark?

   66, 30, 56, 20, 13, 56, 53, 70, 50, 30, 56, 45

3. The weights of 25 students are given below.

   Find the Median Weight?

Weight in Kgs	No.of students
35	4
40	5
50	6
55	6
60	2
65	2

4. If the Median of the numbers 2, 5, 7, 10, 12, x, 17, 19, 21, 24 is 13.

   Find the valur of x?

5. Below is the list of children in a class sorted according to their height.

Height (cm)	Number of Children
145 - 150	6
150 - 155	12
155 - 160	2
160 - 165	5
165 - 170	11
170 - 175	9

a) If the children are placed in order of height,
What is the position of the chid who has median height?

b) What is the estimated height of the 21st student?

c) Calculate the Median height?

6. In a locality the houses are classified according to the consumption of electricity.

Consumption of Electricity	No.of Houses
0 - 60	4
60 - 120	10
120 - 180	12
180 - 240	15
240 - 300	14
300 - 360	4

a) According to the hypothesis what is the consumption of electricity of 27th house?

b) Find the Median?

7. The table shows the ages of 100 peoples.

Age	Number of People
0 - 10	5
10 -20	15
20-30	20
30-40	25
40-50	15
50-60	11
60-70	9
Total	100

a) If the peoples are arranged inorder of their ages, what is the age of 41st person?

b) Find the Median age?

Chapter 13

STATISTICS

ANSWERS

1. a) Mean = 31

   b) Median = 31

2. 13, 20, 30, 30, 45, 50, 53, 56, 56, 56, 66, 70

   Median = 51.5

3. Median Weight $\frac{25+1}{2}$ = Weight of 13th student

   = 50kg

4. Median = $\frac{12+x}{2}$

   ie, $\frac{12+x}{2}=13$

   $x=26-12=14$

5.

Height	Number of Children
Below 150	6
Below 155	12
Below 160	20
Below 165	25
Below 170	36
Below 175	45

   a) 23

   b) 160.5

   c) 162.5

6.

Consumption of Electricity	No.of Houses
below 6	4
below 120	14
below 180	26
below 240	41
below 300	55
below 360	59

a) 182

b) Median = 194

7.

Age	Number of People
below 10	5
below 20	20
below 30	40
below 40	65
below 50	80
below 60	91
below 70	100

a) $30+\frac{5}{25}=30.2$

b) Median = 34

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